



Beyond the horizon: A comprehensive analysis of artificial intelligence-based weather forecasting models

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ABSTRACT

The field of Artificial Intelligence (AI)-based weather forecasting is growing rapidly, with continuous progress in model development, techniques, and performance improvements. This paper provides a comprehensive overview of AI-based weather forecasting models, focusing on their current status, challenges, and directions for further development. A review of more than 40 models, primarily proposed after 2015, underscores the importance of critically examining various aspects of AI-based forecasting. Unlike previous reviews that targeted only a limited number of models or features, this study addresses a complete set of aspects and analyzes existing challenges from multiple perspectives. These aspects include the Machine Learning (ML) and Deep Learning (DL) methods used, datasets, predictand parameters, overfitting, and capability for forecasting extreme weather, lead time, spatiotemporal scale, performance criteria, overfitting, data assimilation, data-driven models, and the analysis of state-of-the-art (SOTA) models such as FengWu, ClimaX, Pangu-Weather, FourCastNet, GraphCast, GenCast, and Artificial Intelligence Forecasting System (AIFS) from various viewpoints. The review also discusses current challenges, including limited historical data and data quality, small-scale weather forecasting, model explainability, uncertainty, extreme weather prediction, physical constraints, temporal adaptation, and generalization, and outlines potential future directions.

1. Introduction

The development of weather prediction has undergone noticeable progress since the 1950s, when computers were first used to mimic the atmosphere's state and changes. Numerical Weather Prediction (NWP) models, which use non-linear partial differential equations called primitive equations to approximate the complicated interaction of mass, momentum, and thermal energy in the atmosphere, were developed (Kimura, 2002). These models start simulations with an analysis that is done using observed data. NWP models represent the evolution of the atmosphere over the Earth, discretized in a grid in three dimensions of space and time.

There are several challenges that hinder NWP models (Abraham et al., 2002). First, due to the chaotic nature of atmospheric phenomena (Abraham et al., 2002), slight variations in initial conditions can substantially influence model outputs. The measurement of initial conditions that are input into the model, data assimilation, and an incomplete

understanding of the atmospheric physical processes all inevitably lead to errors. As time goes on, the precision of prediction reduces with a growing gap between current time and predicted time. Secondly, an outpouring of meteorological data is now available. For example, sensors and autonomous observing platforms have collected petabytes of meteorological data (Agapiou, 2017). In contrast, NWP models generate several terabytes of simulation results daily. Additionally, various types of uncertainties are present in the datasets, and there are spatiotemporal correlations between datasets that pose unprecedented problems to NWP. Thirdly, there is a substantial computation cost associated with numerical solutions to theory-based nonlinear equations, which is mainly dependent on supercomputer capability. Last but not least is the fact that grid resolutions are smaller than certain atmospheric processes' natural scales, typically given in kilometers. However, convection, radiative transfer, and cloud formation at small scales are not explicitly solved but treated with parameterization (Milton and Wilson, 1996; Delage, 1997).

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Artificial Intelligence (AI) is particularly prominent in overcoming challenges across diverse domains, especially post-processing tools for correcting systematic errors in NWP output data, enhancing resolution, and accounting for topographic effects (Aznarte and Siebert, 2017; Buehner et al., 2010). Besides, AI provides an opportunity to assess predictability considering the uncertainty of ensemble forecasting (Wilks, 2002; Foley et al., 2012; Mallet et al., 2009) and addressing problems related to extreme events such as hailstorms, gale storms, or cyclones (McGovern et al., 2017; Williams et al., 2008; Herman and Schumacher, 2018). Although innovative methodologies are developing, NWP simulations remain relevant, and their impact on improving output quality is significant because AI-based models work based on the input data (Wilks, 1995). All these issues, combined with the availability of petabyte-scale climate and weather data on a global scale and the recent reduction in computational demands for Machine Learning (ML) and Deep Learning (DL) models, have sparked interest in integrating AI-based methods into weather and climate modeling among weather scientists and AI researchers. Many of ML algorithms that are used in the core of AI-based weather forecasting models help in correcting the systematic errors that arise out of NWP outputs and improve spatial and temporal resolution. AI does not replace conventional models but enhances them with the ability to model the complexities governing atmospheric dynamics.

Several recent review papers have explored the intersection of AI and weather forecasting, each contributing valuable insights but with notable limitations in scope and coverage. Camps-Valls et al. (2025) focus on the use of AI for modeling and understanding extreme weather and climate events, emphasizing explainability, ethical concerns, and stakeholder engagement. Waqas et al. (2024) provide a technically detailed survey on integrating AI into NWP systems, particularly in areas such as data assimilation, and post-processing. Similarly, Zhang et al. (2025) present a taxonomy of ML techniques and discuss their general applicability in weather forecasting. While informative, these reviews tend to concentrate on specific aspects or applications without delivering a comprehensive, systematic, and comparative evaluation of recent AI-based weather forecasting models. Notably, they do not analyze state-of-the-art (SOTA) models such as FengWu, ClimaX, Pangu-Weather, FourCastNet, GraphCast, GenCast, and the Artificial Intelligence Forecasting System (AIFS), nor assess their performance across real-world forecasting scenarios. In contrast, this review provides a broad analysis of more than 40 AI-based weather forecasting models developed primarily after 2015. It offers a unified examination of various critical aspects, including model architectures, training and evaluation datasets, predictand parameters, extreme event forecasting, spatiotemporal scales, lead times, performance metrics, overfitting, data

assimilation methods, data-driven models, and an analysis of state-of-the-art (SOTA) AI-based weather forecasting models. This provides readers with a comprehensive understanding of the current state and challenges of AI-based weather forecasting, while also offering practical directions for future research. Fig. 1 shows the topics discussed in this review.

Models were selected through a structured search of academic databases using targeted keywords to ensure relevance, innovation, and methodological rigor. The review fills an important gap in the literature by combining both technical and applied perspectives, making it a valuable resource for researchers, practitioners, and decision-makers. A key contribution of this work is its in-depth historical overview, which traces the evolution of AI models in weather forecasting from their early foundations to the most advanced modern techniques. This progression highlights how methodologies have grown in complexity and capability over time, offering essential context for understanding current developments.

2. ML models

This section focuses on the different ML and DL methods used in AI-based weather forecasting models. Depending on their complexity and sophistication, these models are categorized into four classes including traditional ML models, general Neural Networks (NNs), modern DL approaches, and Graph NNs (GNNs).

2.1. Traditional ML methods

Studying the advantages and disadvantages of traditional models used in AI-based weather forecasting research starts with Standard Regression (SR) methods, especially Linear Regression (LR), which is the simplest and most understandable method, but it may not handle non-linear characteristics in the data well (Kuligowski and Barros, 1998; McCrear et al., 2005; Hewage et al., 2021). Kernel Ridge Regression (KRR) is the next method that is a regularization extension intended for regression tasks with both positive and negative aspects when dealing with highly non-linear data sets (Vovk, 2013; Lagomarsino-Oneto et al., 2023). As for K-Nearest Neighbors (KNN), it is known for its simplicity and natural ability to capture localized, changing relationships. However, computational demands are high, especially with larger datasets, while interpretability is limited (Sharma et al., 2014; Taunk et al., 2019). Random Forest (RF) provides a better solution than others that handle non-linearity and can tell us which predictors are important in prediction tasks. Despite this advantage, there are several challenges, including the existence of highly imbalanced data sets as well as

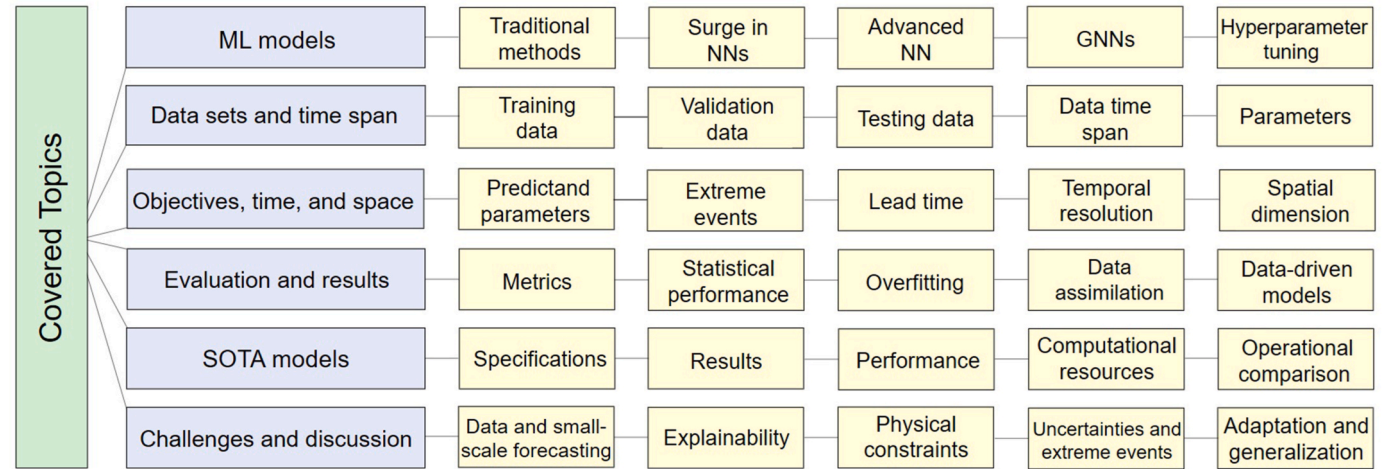


Fig. 1. Overview of the topics covered in this review.

memory-intensive operations that must be computed (Cutler et al., 2012; Hanoon et al., 2021; Hewage et al., 2021). To increase the accuracy, the direction of exploration for improved prediction went towards Gradient Boosting Trees (GBT), which employ a combination of weak learners to create powerful models (Deng et al., 2021; Hanoon et al., 2021). However, GBTs are known to be very sensitive to overfitting, and they also take longer to train. Natural Gradient Boosting (NGBoost) introduces a probabilistic framework to continue the journey through a probabilistic framework that handles uncertainty and improves upon GBT (Duan et al., 2020; Peng et al., 2020). Nonetheless, this advancement has its downside in that it makes computational requirements high. Researchers also turned to using support vector-based approaches, including Support Vector Regression (SVR) and Support Vector Machine (SVM). These approaches are effective in high-dimensional spaces and introduce non-linearity by using kernel tricks (Radhika and Shashi, 2009; Liu et al., 2015; Zhang et al., 2017; Hewage et al., 2021; Lopez et al., 2022; Yin et al., 2022). However, it still has some weaknesses, such as being very sensitive to the choice of kernel type. In capturing complex relationships, a Gaussian kernel SVM (GSVM) utilizes a Gaussian kernel. While capable of handling complex patterns, GSVMs may, however, be prone to overfitting, which means careful parameter setting is vital for their success (Yang et al., 2010; Wei, 2012). On the way to advanced Wavelet kernel SVMs (WSVMs), wavelet kernels are used to achieve a better representation of signal data through the integration of wavelet kernels (Zhang et al., 2004; Wei, 2012). While offering advancements, WSVMs may demand expertise in signal processing and present increased complexity in implementation. The performance of WSVMs can be sensitive to the choice of wavelet functions and other parameters, requiring careful tuning to achieve optimal results. Fig. 2 shows different kinds of traditional ML methods used in previous studies. It should be noted that some of the previous research used different kinds of ML approaches, including traditional and more advanced methods, for comparison. Therefore, those papers will be mentioned in all the related categories.

2.2. Surge in NNs

In recent years, researchers have resorted to NNs as a way of circumventing the limitations of traditional models. NNs are well-suited for parallel computation and adept at handling complex relationships. They have good scalability with regard to computational resources especially when dealing with big data unlike many other traditional models that suffer from computational bottlenecks as data size increases. In this paper, general NNs have been classified into four groups namely shallow NN, multilayer and deep NN, basic Convolutional NNs (CNNs), and basic Recurrent NNs (RNNs). Shallow NNs can be trained quickly and implemented easily. Hence, they cannot capture intricate relationships of complex data that could be used in high-level feature abstraction tasks. In comparison with traditional architectures, a multi-layer model is more powerful in its ability to learn complex non-linear relations.

These kinds of NN models that were used in previous studies include Multi-Layer Perceptron (MLP) and Deep Belief Network for Precipitation Forecast (DBNPF). A major disadvantage of MLPs is that they have a shallow architecture, which consists of only a few hidden layers (Ramchoun et al., 2016). This means they cannot learn and represent detailed hierarchical features in data automatically. In the context of DBNPF, the architecture involves the use of Restricted Boltzmann Machines (RBMs) for unsupervised pre-training and a top layer with backpropagation (Zhang et al., 2017). The incorporation of RBMs and the specific training approach make it different from a traditional MLP. Deep NNs like DBNPF that have deeper architectures are good at capturing complicated interdependencies and getting out higher-order properties like these. However, they tend to overfit and demand intense computational resources for training. More specifically, the problem of needing large amounts of data or inputs, possible vanishing or exploding gradients during training, and increased computational complexity poses some challenges for them. Therefore, deep NNs are generally used at the heart of special NN models to optimal use of them. The CNNs try to capture spatial hierarchies and variability in the input data (Taye, 2023). On the other hand, RNNs suit sequential data and time-series forecasting more appropriately. Still, these models face issues such as vanishing gradient problems as well as computational intensity, which is unparallelizable (Hochreiter and Schmidhuber, 1997; Graves and Schmidhuber, 2005). In Table .1, the techniques for each class including their specifications and contributions to weather forecasting models, are given, respectively.

2.3. Advanced NN methods

Advanced NN methods are diverse and tailored to address specific challenges across various applications. Among the more sophisticated CNN methods, U-shaped network (U-Net), originally designed for biomedical image segmentation, utilizes a specialized architecture to achieve precise segmentation tasks. Segmented NN (SegNet), on the other hand, focuses on semantic segmentation, employing an encoder-decoder architecture with skip connections to preserve spatial information effectively (Larraondo et al., 2019). Temporal Convolutional Network (TCN) is designed for sequential data processing, capturing long-range dependencies efficiently (Hewage et al., 2021). Capsule Networks (CapsNets) introduce capsules as fundamental units, excelling in tasks involving hierarchical relationships (Chattopadhyay et al., 2020). Lastly, Residual Networks (ResNets) mitigate the vanishing gradient problem through skip connections, enabling the construction of deeper network architectures (Nguyen et al., 2023a, 2023b). In the realm of RNNs, advanced methods offer enhanced capabilities for sequence modeling and analysis. Bidirectional LSTM (Bi-LSTM) captures contextual information from both past and future sequences, facilitating a more comprehensive understanding of temporal dynamics (Hewage et al., 2021). Spatiotemporal Stacked-LSTM (SS-LSTM) specializes in analyzing spatiotemporal data, effectively modeling complex temporal

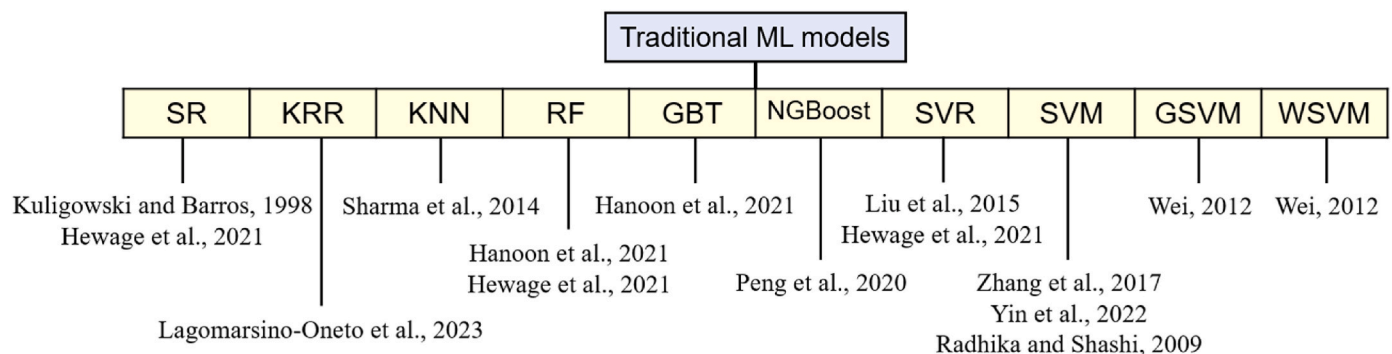


Fig. 2. Traditional ML models used in previous studies.

Table 1
Specifications of different basic NN methods.

Category	Methods	Specifications
Shallow NN	ELM (Extreme Learning Machine)	• Fast learning speed due to a single-layer feedforward network • Limited ability to handle complex tasks compared to deeper models
	RBFN (Radial Basis Function Network)	• Good at capturing non-linear relationships in data • Limited scalability for high-dimensional data
Multi-Layer and Deep NN	MLP	• Versatile and effective for various applications • Prone to overfitting, especially with limited data
	Multi-Task-MLP (MT-MLP)	• Leverages correlations between multiple sites for improved predictions • Increased complexity and potential overfitting with multiple tasks
	DBNPF	• The DBN consists of multiple layers of RBMs • RBMs are used for unsupervised pre-training, capturing hierarchical features • Combines the strengths of RBMs for unsupervised learning and backpropagation for supervised fine-tuning • Accelerates computation speed of network node probability by using RBMs • Helps overcome issues related to weight and bias adjustments when the number of layers increases • Tuning the entire system, including RBM layers, backpropagation, and potential particle filter parameters, may be challenging and time-consuming • Computationally intensive
	CNN	• Excellent at capturing spatial patterns in data • Computationally intensive and requires substantial data
Basic CNN	Multi-Task-CNN (MT-CNN)	• Utilizes correlations between multiple sites for improved predictions • Increased complexity and potential overfitting with multiple tasks
	CNN-LSTM	• Captures both spatial and temporal patterns in data • Requires careful tuning of hyperparameters
	Visual Graphics Group (VGG)-16	• Deep architecture captures complex features • Computationally expensive and may suffer from vanishing gradients
	RNN	• Sequential processing suitable for time-series data • Struggles with capturing long-term dependencies
Basic RNN	LSTM	• Addressing vanishing gradient problem making them well-suited for tasks involving sequences with long-term dependencies • Computationally expensive
	Multi-Task-RNN (MT-RNN)	• Leverages correlations between multiple sites for improved predictions • Increased complexity and potential overfitting with multiple tasks
	Multiple Input, Multiple Output (MIMO)	• Simultaneously predicts multiple outputs • Increases model complexity, challenges in data alignment, and the need for sufficient data
	Multiple Input, Single Output (MISO)	• Simplifies prediction tasks with a single output • Limitations in representing individual aspects separately, challenges in information fusion, and the dependency on proper feature extraction

dynamics within spatial contexts (Karevan and Suykens, 2018). Fully Convolutional LSTM (FC-LSTM) extends convolutional operations to the temporal domain, allowing for end-to-end learning (Shi et al., 2015). Transformer-based architectures have revolutionized sequence processing tasks, leading to significant advancements in various domains. Vision Transformer (ViT) uses self-attention mechanisms for sequence data (Nguyen et al., 2023a, 2023b). Three-Dimensional (3D) NNs are tailored for volumetric data analysis, capturing spatial dependencies within 3D data (Bi et al., 2023). Shifted Window transformer-based Recurrent NN (SwinRNN) integrates transformer blocks into recurrent architectures, enhancing sequential modeling capabilities. Swin transformer-based Variational RNN (SwinVRNN) extends SwinRNN to incorporate variational inference, enabling probabilistic modeling of sequential data (Hu et al., 2023). Adaptive Fourier Neural Operator (AFNO) leverages Fourier analysis to learn adaptive basis functions for efficient representation of complex data patterns (Pathak et al., 2022). Cross-Modal Fusion Transformer (CMFT) is designed for multimodal data fusion, effectively integrating information from diverse sources through attention mechanisms (Chen et al., 2023b). Autoencoder methods offer unsupervised learning techniques for feature extraction and data compression. Sparse AutoEncoder model (SAE) introduces sparsity constraints to encourage the learning of robust and compact representations (Zhang et al., 2017). Autoencoders-based Deep NN with SVR (SVRDNN) incorporate SVR into deep autoencoder architectures for enhanced performance in regression tasks (Liu et al., 2015). Generative methods, including Variational AutoEncoders (VAEs) and diffusion models, enable the probabilistic generation of data samples. VAEs are a type of autoencoder that incorporates probabilistic techniques for generating data. Unlike traditional autoencoders, which learn deterministic mappings from input to latent space, VAEs learn a probabilistic distribution over latent space (Chen et al., 2023c; Kwok and Qi, 2021). Diffusion models employ iterative diffusion processes to model data distribution effectively, generating realistic and diverse samples. In a diffusion model, the initial data sample is gradually transformed into a new sample through a series of diffusion steps. At each step, noise is added to the data sample, leading to a diffusion process that gradually spreads the data distribution (Hua et al., 2024; Gao et al., 2024). Table .2 shows the specifications and contribution of various classes of advanced NN methods.

Fig. 3 provides a comprehensive overview of the basic and advanced NNs used in AI-based weather forecasting models.

Weather forecasting involves analyzing and predicting complex, nonlinear, and high-dimensional atmospheric systems. To effectively address this complexity, data-driven techniques such as Proper Orthogonal Decomposition (POD) and Dynamic Mode Decomposition (DMD) are frequently employed (Manning and Baldick, 2019; Sharma et al., 2024; Teixeira et al., 2025). POD identifies dominant spatial structures such as pressure systems, temperature fields, and wind patterns, allowing the system to be represented in a reduced-order form. This preserves key features while substantially lowering computational costs, enabling faster and more efficient simulations (My Ha et al., 2008; Wang et al., 2023). DMD captures the temporal evolution of coherent atmospheric structures by extracting dominant dynamic modes along with their growth or decay rates. Operating in a model-free and data-driven manner, DMD can be applied directly to observational or simulation data, making it especially valuable for analyzing transient phenomena such as atmospheric blocking, wave propagation, and storm development (Mansouri et al., 2023; Sofi and Oseledets, 2024; Palma et al., 2024). By reducing data complexity and isolating essential patterns, POD and DMD enhance the efficiency and accuracy of subsequent learning processes when integrated with traditional or DL models.

2.4. Unraveling the potential of GNNs

One major issue overlooked in current NN architectures is that they cannot accurately describe the complex spatial relations between

Table 2
Specifications of advanced NN methods.

Category	Methods	Specifications
Advanced CNN	U-Net	• Excels even with limited annotated data, making it suitable for scenarios where large labeled datasets are challenging to obtain • The inclusion of skip connections enhances its performance by facilitating the flow of high-resolution features, aiding in accurate object localization • May have limitations in grasping extensive global context and may struggle with capturing long-range dependencies
	SegNet	• Facilitates hierarchical feature extraction through its encoding path, enabling the capture of both low-level and high-level features for semantic segmentation. The decoding path's upsampling strategy effectively restores the spatial information lost during the encoding phase, aiding in accurate pixel-wise classification • Complexity and potential overfitting in non-segmentation tasks
	TCN	• Captures long-range dependencies efficiently • May require careful tuning of hyperparameters • TCN's performance can be sensitive to the length of input sequences, and careful consideration is required when dealing with very short or very long sequences
	CapsNets	• Use dynamic routing, allowing capsules in different layers to communicate and vote on the presence of entities in the input data • The dynamic routing mechanism helps in capturing hierarchical relationships between features • CapsNets may be less prone to overfitting compared to traditional CNNs, as they learn to recognize entities based on their instantiation parameters rather than fixed features. • Computationally intensive dynamic routing mechanism
	ResNets	• Addresses vanishing gradient problem with skip connections • Residual learning simplifies the optimization process by allowing the network to learn residual functions rather than the entire mapping • ResNets, especially very deep ones, may be prone to overfitting, especially when dealing with small datasets
Advanced RNN	Bi-LSTM	• Captures information from both past and future time steps • Increases computational complexity compared to unidirectional LSTMs • Like other DL models, may be susceptible to overfitting
	SS-LSTM	• Captures both spatial and temporal dependencies • Computational complexity, potential overfitting
	FC-LSTM	• Efficiently captures spatiotemporal patterns • Requires careful tuning of hyperparameters
Transformer	Transformer	• Parallel processing of input sequences, making transformers computationally efficient • Transformers can scale to handle sequences of varying lengths without the need for recurrence • Training large transformers can be computationally intensive, requiring significant resources • Probable need for large datasets and may be prone to overfitting
	ViT	• Effectively applies self-attention mechanisms for sequence data • ViT's reliance on patches and positional encodings may limit its ability to understand fine-grained spatial information • Computationally intensive for large datasets

Table 2 (continued)

Category	Methods	Specifications
	3D NN	• Standard with encoder–decoder design of ViTs • Integrates 3D information for improved modeling • Increased computational demands
	SwinRNN	• Efficiently captures spatiotemporal dependencies • Requires careful tuning of hyperparameters
	SwinVRNN	• Integrates variational techniques for uncertainty estimation • Combines a deterministic SwinRNN predictor with a perturbation module to introduce stochasticity • Increased complexity and potential computational demands
	U-Transformer	• Influenced by the efficiency of the Swin Transformer architecture • Simultaneously process sequential input data, capturing dependencies across different elements of the sequence May require substantial computational resources
	AFNO	• Combines the principles of the Fourier Neural Operator learning approach with the power of a ViT • This integration likely aims to leverage the strengths of Fourier analysis in capturing frequency information and ViT's ability to model complex relationships • May result in increased computational demands and may require careful tuning of hyperparameters
	CMFT	• Integration of information from different modalities seamlessly • Potentially learn more robust and comprehensive representations compared to unimodal models • Adapted to various multimodal tasks • Training CMFT models typically requires large amounts of labeled multimodal data • Aligning semantic information across different modalities can be challenging
	Autoencoder	• Easy to implement and understand • Limited capability for capturing complex patterns
	SAE	• Can automatically learn relevant features from data without manual feature engineering • Sensitive to hyperparameter choices and may require careful tuning
	Auto-MLP	• Streamlines the feature extraction process, leveraging the strengths of both autoencoders and MLPs • May require careful tuning of hyperparameters, and the performance is contingent on the quality of learned features.
	SVRDNN	• Hierarchical feature learning through stacked Autoencoders • Integration of SVR for regression tasks • Potential for capturing complex patterns in data • Requires significant computational resources and training time
Generative Methods	VAE	• Sensitivity to hyperparameter choices and risk of overfitting • Learns a latent representation for data generation and feature extraction • Minimizing the Kullback-Leibler (KL) divergence loss in addition to the reconstruction error in training step • Follows a specific distribution like normal • More complex training than other autoencoders
	Diffusion model	• Generate new data samples resembling the training data • Gradually add noise to data, then learn to reverse the process • Can utilize various architectures like CNNs, RNNs, or Transformers for denoising

(continued on next page)

Table 2 (continued)

Category	Methods	Specifications
		•Can be computationally expensive, training is sensitive to hyperparameters

features of weather data. Such complex regional patterns are difficult to model with grid-based representations, hence predictions get less and less accurate. Also, it may be difficult in these kinds of models to build time dependencies into their architectures. This is where GNNs emerge as a promising solution for AI-based weather forecasting. GNNs are meant for handling data with natural graph structures, which makes them very good at representing and interpreting complex spatial relationships within weather data (Wu et al., 2022). Unlike grid-based models, GNNs excellently capture dependencies among geographically distributed nodes, such as particular sites or weather stations. GNNs have the distinct advantage of modeling both spatial and temporal dependencies simultaneously (Liang et al., 2022). Unlike RNNs and CNNs, that might fail to capture fine-grained spatiotemporal patterns and may lack a natural way of including temporal dependencies, respectively. Another reason is that GNNs are applicable to irregular graph structures, which is the case in weather forecasting (Lin et al., 2022). This suggests that GNNs have a broader application scope in weather forecasting where meteorological conditions are not confined to a regular grid. This means that GNNs do not just adopt a universal format for each element but instead focus on how they are structured. GNNs use the message-passing approach, which lets nodes extract information from nearby nodes and detect local phenomena that should be taken into account for accurate predictions. Different GNN models implement the encoding, processing, and decoding of weather-related information in different ways within the realm of AI-based weather forecasting. Keisler (2022) proposed a model based on an encode-process-decode framework with specific graph structures for each step. To utilize an encoded input data set from a latitude/longitude grid, this bipartite-graph system extracts out abstract latent features on an icosahedron mesh construction. The latent features are processed iteratively by message-passing GNNs in its processor in order to predict 6-h changes in atmospheric variables by its decoder, which finally maps back to physical data on a latitude/longitude grid. On the other hand, GraphCast model, which was developed in 2023 by Lam et al., utilizes multi-mesh graph

representations for global spatial modeling. It uses GNNs with an encode-process-decode arrangement that allows information to flow between grid and mesh nodes. For this purpose, the encoder encodes a latent space where features of different graph components are represented. Meanwhile, the processor uses a multi-layer interaction network to facilitate message passing within a mesh subgraph. The decoder carries out message passing over Mesh2Grid's subgraph so as to predict each grid node using an output MLP. The model used by Keisler (2022) focuses on an icosahedral grid and particular atmospheric variables for its own domain-specific applications, whereas GraphCast may have some advantages when it comes to capturing long-range dependencies using its multi-mesh approach and iterative processing. Other GNN-based methods used in spatiotemporal forecasting include Conditional Local Convolution Recurrent Network (CLCRN), Time domain Graph Convolutional Network (GCN) (TGCN), Multi-view Spatiotemporal GCN (MSTGCN), Attention-Based Spatiotemporal GCN (ASTGCN), Graph Convolutional Gated Recurrent Unit (GCGRU), Diffusion Convolutional RNN (DCRNN), and Adaptive Graph Convolutional Recurrent Network (AGCRN) (Yu et al., 2018; Bai et al., 2020; Lin et al., 2022). Table 3 shows the contribution of GNN structures in AI-based weather forecasting models. Sophisticated GNN architectures, attention mechanisms, and temporal modeling techniques are incorporated into these models. For instance, CLCRN combines graph-based spatial representation with conditional local convolution and recurrent layers to capture spatial and temporal dependencies. On the other hand, TGCN uses graph convolutional networks in the time domain to capture spatial and temporal aspects, while MSTGCN integrates multi-view data and spatio-temporal GCNs. Attention mechanisms for improved modeling are incorporated in ASTGCN, while GCGRU combines graph convolutional networks with gated recurrent units. DCRNN values diffusion convolutional layers within a RNN framework as its unique distinguishing feature. Moreover, DCRNN effectively captures spatial dependencies by leveraging information diffusion across the graph so that it can grasp long-range interactions in meteorological data. Conversely, AGCRN introduces adaptability by dynamically changing the structure of the graph during training. This dynamic adaptation enables the model to capture evolving relationships in spatiotemporal data. Unlike others, AGCRN is able to modify its graph connections flexibly through parameter learning during trainings so as to change its own dependency structure over space (Lin et al., 2022). Recently, Lang et al. (2024)

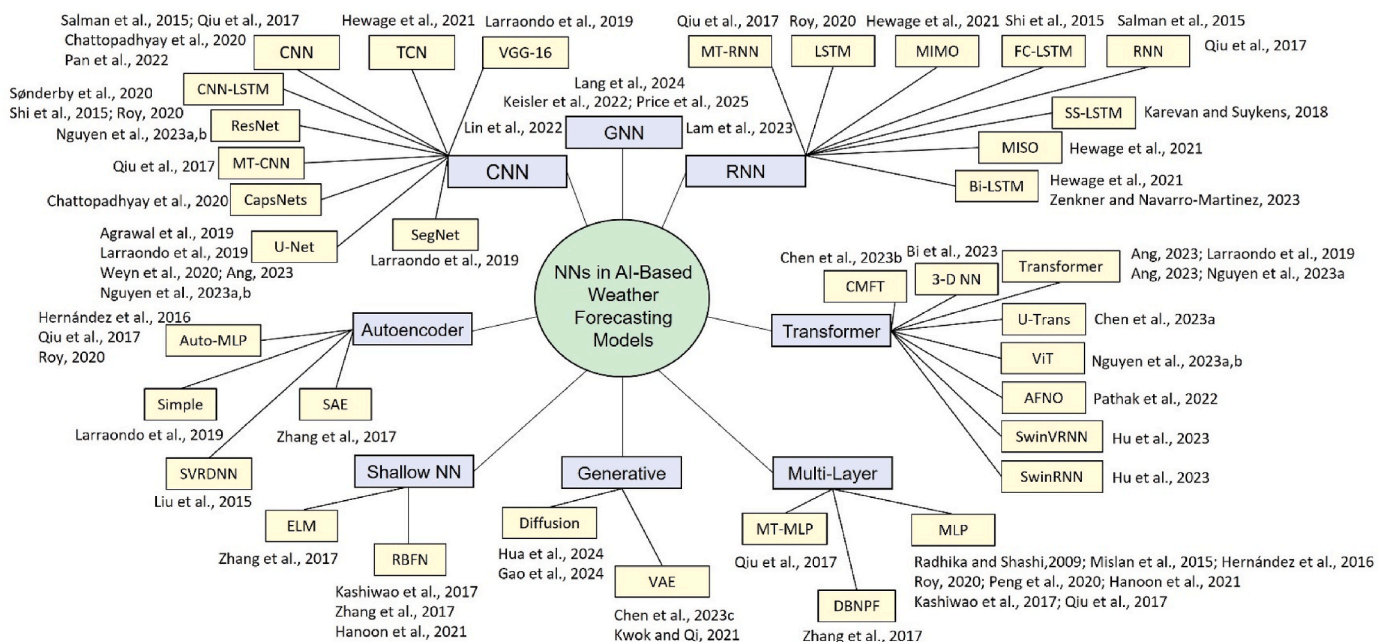


Fig. 3. Contribution of NNs in AI-based weather forecasting models.

Table 3

Contribution of GNNs in AI-based weather forecasting models.

GNNs in AI-Based Models		Specifications
Keisler, 2022		• The Encoder transforms data from a grid to abstract latent features on an icosahedron grid using a bipartite graph • Initial node features include atmospheric variables, solar radiation, orography, land-sea mask, day-of-year, latitude, and longitude • Edge features involve the positions of latitude/longitude nodes relative to each icosahedron node • The Processor processes the data on the icosahedron grid through GNNs • The Decoder then maps these abstract features back to physical data on a latitude/longitude grid
Lin et al. (2022)	CLCRN	• Graph-based spatial representation + spatial convolution + conditional local convolution with recurrent layers + meteorological data • Introduces conditional local convolution for mimicking meteorological flows + spatial convolution for local pattern capture
	TGCN	• Using a GCN to capture temporal dependencies in spatial data considering both spatial and temporal aspects in a unified framework
	MSTGCN	• Multi-view data + spatiotemporal graph convolutional networks • Handles diverse data views, such as different types of meteorological data, for an understanding of interactions
	ASTGCN	• Attention mechanisms + spatiotemporal GCNs • Utilizes attention mechanisms to weigh the significance of nodes and edges in the graph, improving the modeling process
	GCGRU	• GCN + Gated recurrent units • Introduces GRUs within the graph convolutional framework to capture both spatial and temporal features effectively
	DCRNN	• Diffusion convolutional layers + RNN • Introduces diffusion convolutional layers to propagate information across the graph to capture long-range dependencies
	AGCRN	• Adapts graph structure dynamically during training + captures evolving relationships in spatiotemporal data • Dynamically adjusts graph connections based on learned parameters, enabling the model to adapt to changing spatial dependencies
		• Graph representation with grid nodes for atmospheric slices and mesh nodes forming a refined global mesh • Bidirectional mesh edges and unidirectional Grid2Mesh edges and Mesh2Grid edges enable data exchange between grid and mesh nodes • The Encoder embeds features of grid nodes, mesh nodes, mesh edges, Grid2Mesh edges, and Mesh2Grid edges into a latent space • Conducts message passing from grid nodes to mesh nodes, facilitating the transfer of atmospheric state information • The Processor involves a deep GNN operating on the mesh subgraph, incorporating mesh nodes and mesh edges • The Processor employs a multi-layer interaction network for message passing and operates on a multi-mesh • The Decoder uses a Mesh2Grid GNN to perform message passing over the Mesh2Grid subgraph, transmitting information back to grid nodes • The Decoder subsequently updates grid nodes based on the aggregated information obtained from Mesh2Grid edges
Lam et al. (2023) (GraphCast)		
Lang et al. (2024) (AIFS)		• Instead of simple message passing, these models use attention mechanisms (from transformers) to compute how much each node should “attend” to or weigh information from other nodes. • Each node’s new representation is a weighted sum of all or a subset of other nodes’ features, where weights (attention scores) are learned dynamically

Table 3 (continued)

GNNs in AI-Based Models	Specifications
Keisler, 2022	• The Encoder transforms data from a grid to abstract latent features on an icosahedron grid using a bipartite graph • Initial node features include atmospheric variables, solar radiation, orography, land-sea mask, day-of-year, latitude, and longitude • Edge features involve the positions of latitude/longitude nodes relative to each icosahedron node • The Processor processes the data on the icosahedron grid through GNNs • The Decoder then maps these abstract features back to physical data on a latitude/longitude grid
Price et al. (2025) (GenCast)	• This is based on the shifted window attention which is a transformer technique that computes attention in overlapping local windows, enabling efficient, scalable attention while capturing neighborhood information • Uses a diffusion probabilistic model with a graph-transformer-based denoiser architecture • Graph-transformer is a special kind of GNN that uses transformer-style self-attention instead of (or in addition to) traditional message passing. • Instead of just aggregating neighbors’ features uniformly or by simple weights, the transformer attention mechanism computes dynamic attention weights • Operating on a spherical mesh to autoregressively forecast future atmospheric states conditioned on the previous two states. • The diffusion process is adapted to the spherical geometry of Earth

proposed AIFS, a model that replaces traditional GNNs with a transformer-based architecture incorporating self-attention. Rather than relying on explicit edge features such as length and direction, AIFS uses self-attention across nodes, allowing each node to dynamically learn how much information to draw from others based on learned attention scores. This design removes the need for fixed graph edges and instead structures attention along latitude bands. Within these bands, AIFS applies shifted window attention, enabling nodes to attend to overlapping local regions and effectively capture both local and global spatial dependencies. The key distinction from conventional GNNs lies in the method of information sharing: while traditional GNNs rely on edge-based message passing, AIFS enables broader contextual access through learned attention. Similarly, GenCast, introduced by [Price et al. \(2025\)](#), is also built on a graph-transformer framework. It leverages self-attention to dynamically weigh neighboring nodes on the Earth’s spherical mesh, allowing it to model complex, global weather patterns more effectively than standard GNNs.

2.5. Hyperparameter tuning

Hyperparameter tuning is a crucial component in the development and optimization of AI-based weather forecasting models. They greatly influence the learning dynamics, convergence behavior, and ultimately the forecasting accuracy of a model. The correct configuration of these parameters can significantly enhance a model’s generalization ability, reduce overfitting, and improve performance on unseen weather data, which is particularly critical given the high variability and complexity of atmospheric systems. [Table .4](#) presents a comparative summary of hyperparameters and tuning strategies employed across a wide range of studies. The table highlights a noticeable variation in the selection and tuning of hyperparameters, reflecting the diversity of model architectures and forecasting tasks. While some studies, such as those by [Kuli-gowski and Barros \(1998\)](#) and [Hernández et al. \(2016\)](#), employed systematic methods like sensitivity analysis and grid search, many others either did not report tuning strategies or relied on empirical or manual selection. Notably, more recent studies (e.g., [Larraondo et al.](#),

Table 4
Hyperparameters and tuning methods.

Study	Mentioned hyperparameters in the text	Tuning method
Kuligowski and Barros (1998)	Number of input nodes, number of hidden-layer, learning rate, momentum term, number of training epochs	Sensitivity, cross-validation, and empirical heuristics
Radhika and Shashi (2009)	Kernel parameters	Grid search
Wei (2012)	Kernel parameters	–
Sharma et al. (2014)	Number of nearest neighbors	–
Liu et al. (2015)	Weight decay parameter, sparsity parameter, regularization parameter, and kernel function	–
Mislan et al. (2015)	Number of neurons, learning rate, number of epochs	–
Salman et al. (2015)	–	–
Shi et al. (2015)	Number of layers, number of hidden states, learning rate, and kernel size	–
Hernández et al. (2016)	Hidden neurons, learning rate, corruption level, iterations, hidden layers, learning rate, and momentum factor	Grid search
Kashiwao et al. (2017)	Hidden layer size, activation function, step sizes, convergence thresholds, and Gaussian spread	–
Qiu et al. (2017)	Regularization weights, kernel sizes, window size, learning rate	–
Zhang et al. (2017)	Number of layers, number of nodes, learning rate, number of iterations, and activation function	Gradient descent
Karevan and Suykens (2018)	Number of neurons per layer, activation function, sequence length	Manual tuning
Larraondo et al. (2019)	Number of epochs, number of input variables, learning rate, and optimizer	Exhaustive search for number of input variables
Chattopadhyay et al. (2020)	Kernel size, regularization constant, and dropout probability	Trial and error on a validation set
Peng et al. (2020)	Number of nodes in hidden layers, activation function, optimization method, learning rate, Base learner type, and number of base learners	Analogous-linear search
Roy (2020)	Number of neurons, activation function, optimizer, and number of epochs	–
Sønderby et al. (2020) (MetNet)	–	–
Weyn et al. (2020)	Filter size, pooling size, number of filters, activation function, optimizer, learning rate, early stopping, number of epochs	–
Hanoon et al. (2021)	Learning rate, number of hidden layers, number of neurons per layer, activation function, number of training epochs, activation function, and number of hidden neurons	–
Hewage et al. (2021)	Number of layers, number of nodes per layer, learning rate, optimizer, filter size, activation function, kernel	Multiple configurations of hyperparameters were tried

Table 4 (continued)

Study	Mentioned hyperparameters in the text	Tuning method
Kwok and Qi (2021)	size, batch size, dilations, and dropout rate Dropout rate, learning rate, learning rate, batch size, number of training epochs, and optimizer	–
Lin et al. (2022)	Kernel parameter, number of epochs, optimizer	Sensitivity analysis
Keisler (2022)	Number of message-passing rounds, number of layers in MLPs, activation function, and optimizer	–
Pan et al. (2022)	Batch size, number of training epochs, and ensemble size	–
Pathak et al. (2022) (FourCastNet)	Patch size, number of layers, token dimension, sparsity threshold, learning rates, batch size, epochs, and cosine learning-rate schedule	Experiments using validation data
Yin et al. (2022)	Tolerance level, width of the radial basis function, and penalty parameter	Random search approach
Ang (2023)	Number of hidden neurons, number of encoder layers, number of decoder layers, number of heads in Multi-Head Attention, and training data duration	–
Bi et al. (2023) (Pangu-Weather)	Base channel width, patch size, number of encoder layers, number of decoder layers, window size, drop ratio, batch size, learning rate, weight decay, loss weights, number of training epochs	–
Lam et al. (2023) (GraphCast)	Number of GNN layers, learning rates, batch size, optimizer, weight decay, weights for variables	Experiments using validation data
Chen et al. (2023a) (FuXi)	Kernel size, stride, number of transformer blocks, and activation function	–
Chen et al. (2023b) (FengWu)	Number of layers, number of attention heads, and hidden dimension size	–
Hu et al. (2023)	Window size, kernel size, stride, number of blocks, number of stages, number of epochs, latent dimension size, and learning rate	–
Lagomarsino-Oneto et al. (2023)	Regularization parameter and kernel width parameter	Grid search using cross-validation
Nguyen et al. (2023a) (ClimateLearn)	Padding size, kernel size, stride, hidden dimension, residual blocks, dropout, and patch size	–
Nguyen et al. (2023b) (ClimaX)	Weight decay, learning rate, and number of training epochs	–
Zenkner and Navarro-Martinez (2023)	Optimizer, learning rate, model optimized metric, number of epochs, batch size, and dropout rate	–
Hua et al. (2024)	Total number of diffusion time steps, diffusion noise scale scheduling, guidance scale for classifier-free guidance, and weighting factor	–
Gao et al. (2024) (PreDiff)	Latent dimension channels, kernel sizes, number of blocks, number of cuboid attention blocks, optimizer, batch size, activation function	–

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Table 4 (continued)

Study	Mentioned hyperparameters in the text	Tuning method
Lang et al. (2024) (AIFS)	Batch size, optimizer, weight decay, learning rate, and total trains steps	–
Price et al. (2025) (GenCast)	Batch size, optimizer, and learning rate	–

2019; Pathak et al., 2022; Lam et al., 2023) used validation-based experiments and more structured search approaches, such as random or exhaustive search, signaling a shift toward more rigorous tuning practices. However, a significant number of models still do not specify tuning methods, raising concerns about reproducibility and performance optimization. The absence of transparent tuning procedures also makes it difficult to assess the fairness and robustness of comparative model evaluations. This inconsistency underscores the need for more standardized hyperparameter tuning frameworks in AI-based weather forecasting research.

3. Datasets and time span

One of the most important factors that affects the accuracy and quality of the final results of both the NWP and AI-based models is the quality of the input dataset. A large amount of meteorological observation data has been collected by several sensors and observing platforms, including those located in space, the ocean, the ground, and the air. Training, validation, and testing datasets and the time span that have been used in current models are shown in Table .5. These data comprise meteorological station observations, NWP models, radar data, and simulation models. The contribution percent of each dataset can be seen in Fig. 4. It should be noted that directly mentioning the details of the models and data is one of the most important aspects of a scientific paper. However, some of the papers didn't clearly mention some of the specifications of the models used.

Meteorological station observations and reanalysis data from the ECMWF, such as ERA-Interim and ERA5, have the highest contribution in current AI-based weather forecasting models. Meteorological station observations are widely used in local models due to their high quality and accuracy (Roy, 2020; Hanoon et al., 2021; Lagomarsino-Oneto et al., 2023; Zenkner and Navarro-Martinez, 2023). In contrast, reanalysis data are generally used in global scale models (Keisler, 2022; Bi et al., 2023; Chen et al., 2023a; Lam et al., 2023). Global coverage with fine spatial resolution, comprehensive atmospheric variables, high temporal resolution, widely accepted standards, accessibility, and providing a long-term and consistent record of past weather conditions are among the common reasons why ECMWF reanalysis datasets are extensively utilized worldwide. Despite these advantages, meteorological station observation data have some challenges that include inherent measurement inaccuracies and discrepancies with numerical models. Data assimilation methods are used to address these problems. Data assimilation could also improve the initial model conditions and correct systematic model biases. It goes beyond individual meteorological models to complex Earth system models that incorporate dynamics from atmospheric, oceanic, and terrestrial subsystems. The reanalysis datasets, such as ERA5, result from assimilation, where the model state is updated (Hersbach and Dee, 2016).

4. Predictand parameters, ensemble forecasting, extreme events, time, and space

4.1. Predictand parameters

Some of the current AI-based models are specialized in predicting a few meteorological parameters (up to 3 parameters), called limited-

variable models, while some others focus on multi-variable forecasting. The multi-variable forecasting models provide a more complete knowledge of weather conditions by admitting the delicate interaction between meteorological parameters. For example the temperature perception affected by increased relative humidity or the pollutant dispersion that is influenced by wind speeds.

4.2. Lead time and temporal resolution

According to their lead time, weather forecasting models can be divided into short-term, medium-term, and long-term forecasts. Short-term forecasts are more focused on the current state of the atmosphere, which includes weather conditions from within an hour up to three days (Frnda et al., 2022). Medium-term predictions are able to go beyond this time frame and generally describe typical weather conditions for a period of three days up to 10 days (Bengtsson, 1984). In contrast, long-term forecasting covers much longer periods, ranging from two weeks to months, years, or even decades. Another important aspect of weather forecasting models is temporal resolution. High temporal resolution enables the models to capture and understand rapid changes in weather variables. This is particularly important for phenomena such as thunderstorms, where conditions can evolve quickly.

4.3. Spatial dimension

Various weather forecasting models focus on at different spatial scales including a city, a country, a continent, or whole of the world. Generally, weather models at the local scale have finer spatial resolutions. Therefore, they can explain how local geography and topography might influence weather patterns in subtle ways. However, most of the local AI-based weather forecasting models focus on meteorological stations that provide low spatial resolution. Global models is based on atmospheric conditions occurring globally, thus having wider coverage but not at finer spatial resolutions as regional models do. Consequently, global forecasts may not capture localized weather events as precisely as regional forecasts. Despite this limitation, global models play a critical role in comprehending large-scale atmospheric dynamics and supplying essential boundary conditions for regional models. Table .6 shows objective parameters and spatiotemporal specifications of the models.

In earlier models, the focus was primarily on limited-variable prediction equivalent up to three variables such as precipitation amount, temperature, and wind speed. However, in recent models, there has been a shift towards predicting a greater number of parameters, indicating a transition towards more comprehensive models. Though extreme weather has started receiving some attention, the capability of most of the current models has not been studied in extreme weather prediction. Nonetheless, some models, such as the one developed by Lam et al. (2023), focus on different types of severe weather situations, including tropical cyclones, atmospheric rivers, and high temperatures. Spatial scale has expanded significantly over time. The earlier models are focused on particular places or stations while the most recent ones include bigger areas or even a global scale. Prediction lead time remains an important feature of weather forecasting. The majority of the possible solutions concentrate mainly on short-term predictions, which usually range from a few hours up to three days; however, there is a growing interest in extending prediction horizons to medium and long-term forecasts, especially due to the emphasis on extreme weather events. Temporal resolution varies across models, with some offering hourly updates while others provide daily or even monthly forecasts.

4.4. Ensemble forecasting and extreme weather

Uncertainties in weather models arise primarily from three sources: inaccuracies in model structure, fluctuations in input data, and the fundamentally chaotic behavior of atmospheric systems. These uncertainties hold particular importance in NWP and climate modeling,

Table 5

Various datasets used in AI-based weather forecasting models.

Study	Training and validation data	Parameter	Testing data	Data time span	Training and validation time span	Testing time span
Kuligowski and Barros (1998)	National Centers for Environmental Prediction (NCEP)	Total column precipitable water, column-average relative humidity, equivalent potential temperature, vertical velocity, zonal/meridional velocity, horizontal divergence, absolute vorticity advection, thermal advection, surface pressure, temperature, geopotential, and absolute vorticity at 1000-, 950-, 850-, 700-, and 500-hPa levels	Two precipitation gauges	5 years	–	–
Radhika and Shashi (2009)	Meteorological station	Daily maximum temperature	Meteorological station	5 years	–	–
Wei (2012)	Central Weather Bureau (CWB)	Pressure at typhoon center, latitude and longitude of typhoon center, radius of typhoon, speed of typhoon, max wind speed at typhoon center, hourly precipitation	Rain gauge data	18 years	–	–
Sharma et al. (2014)	Meteorological station	Station id, year, month, day, temperature, dew Point, wind speed	Meteorological station	4 years	–	–
Liu et al. (2015)	Meteorological station	Temperature and wind speed	Meteorological station	30 years	–	–
Mislan et al. (2015)	Meteorological station	Precipitation	Meteorological station	27 years	22 years	5 years
Salman et al. (2015)	1. Meteorological stations 2. El Nino Southern Oscillation (ENSO)	1. Mean and maximum temperature, minimum temperature, precipitation temperature, relative humidity, mean sea level pressure, mean station pressure, visibility, wind speed, and precipitation 2. Wind, oscillation index, sea surface temperature (SST), and outgoing long wave radiation	Meteorological station	36 years	1–18 years 2–27 years	1–18 years 2–9 years
Shi et al. (2015)	Radar echo data	Precipitation	Radar echo data	3 years	–	–
Hernández et al. (2016)	Meteorological station	Precipitation, temperature, humidity, Pressure, wind speed, and other related variables	Meteorological station	12 years	~10 years	~2 years
Kashiwao et al. (2017)	Meteorological stations	On site and sea level pressure, precipitation, temperature, vapor pressure, humidity, and wind speed	Meteorological station	12 years	11 years	1 year
Qiu et al. (2017)	Meteorological stations	1. Daily accumulated precipitation 2. Wind speed and wind direction, humidity, temperature, precipitation condition, and air pressure	Meteorological station	1. 12 years 2. 3 years	–	–
Zhang et al. (2017)	Meteorological station	Mean pressure, daily average temperature, water pressure, and average relative humidity, small amount of evaporation, average wind speed, and sunshine hours	Meteorological station	54 years	44 years	10 years
Karevan and Suykens (2018)	Meteorological stations	18 measured weather variables, such as temperature and humidity, for each day per city	Meteorological station	7.5 years	7.3 years	2 months
Larraondo et al. (2019)	European Center for Medium-Range Weather Forecasts (ECMWF) (ERA-Interim)	Geopotential and total precipitation at different pressure levels	ECMWF (ERA-Interim)	40 years	–	–
Chattopadhyay et al. (2020)	Large-Ensemble (LENS) community project	Daily-averaged surface temperature, and geopotential at 500 hPa	LENS	85 years	–	–
Peng et al. (2020)	Environmental Modeling Center, Global Ensemble Forecast System (EMC-GEFS)	2-m maximum air temperature	National Oceanic and Atmospheric Administration Climate Prediction Center (NOAACPC)	17 years	–	–
Roy (2020)	Meteorological station	Average wind speed, precipitation, snowfall, snow depth, average temperature, maximum and minimum temperature	Meteorological station	10 years	8 years	2 years
Sønderby et al. (2020) (MetNet)	1. Multi Radar Multi Sensor (MRMS) 2. Geostationary Operational Environmental Satellite 16 (GOES-16)	Precipitation	MRMS and GOES-16	1.5 year	–	–
Weyn et al. (2020)	ECMWF (ERA5)	Geopotential at 500 hPa and at 1000 hPa, 300- to 700-hPa geopotential thickness, and temperature	ECMWF (ERA5)	40 years	38 years	2 years
Hanoon et al. (2021)	Meteorological station	Meteorological data, specifically air temperature, and relative humidity	Meteorological station	33 years	27 years	6 years

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Table 5 (continued)

Study	Training and validation data	Parameter	Testing data	Data time span	Training and validation time span	Testing time span
Hewage et al. (2021)	Weather Research and Forecasting (WRF)	Surface temperature, surface pressure, wind direction, specific humidity, convective rain, non-convective rain, snow water equivalent, soil temperature, and soil moisture	Ground truth data and WRF model	0.5 years	–	–
Kwok and Qi (2021)	Meteosat geostationary meteorological satellites	Cloud top temperature and height, convective precipitation rate, automatic satellite image interpretation - tropopause folding detection, cloud mask, and cloud type	Meteosat geostationary meteorological satellites	2 years	–	–
Lin et al. (2022)	WeatherBench (WB) data (a preprocessed subset of ERA5)	Meteorological parameters including temperature, cloud cover, humidity, and surface wind components	WB data	9 years	–	–
Keisler (2022)	1. ECMWF (ERA5) 2. The Global Forecast System (GFS)	Six physical variables including temperature, geopotential, specific humidity, the eastward component of the wind, the northward component of the wind, and the vertical component of the wind at 13 pressure levels	ECMWF (ERA5), GFS, and Lead Center for Deterministic Forecast Verification (LCDNV)	1. 41 years 2. 0.5 years	1. 38 years 2. -	1. 3 years 3. -
Pan et al. (2022)	Climate simulation data from Coupled Model Intercomparison Project (CMIP5/CMIP6)	Upper ocean potential temperature profile at 8 ocean top layers: 5, 15, 25, 35, 45, 55, 75, and 100 m	Global Precipitation Climatology Project (GPCP) and ECMWF (ERA5)	52,201 years	–	–
Pathak et al. (2022) (FourCastNet)	ECMWF (ERA5)	Various atmospheric variables such as geopotential, temperature, wind speed, and precipitation	ECMWF (ERA5)	40 years	38 years	1 year
Yin et al. (2022)	MesoScale-Model Grid Point Value (MSM-GPV)	Precipitation	Radar-Automated Meteorological Data Acquisition System (Radar-AMeDAS)	13 years	12 years	1 year
Ang (2023)	ECMWF (ERA5)	Cloud cover and precipitation	ECMWF (ERA5)	3.5 years	3 years	0.5 years
Bi et al. (2023) (Pangu-Weather)	ECMWF (ERA5)	69 factors, including 5 upper-air variables at 13 pressure levels (50 hPa, 100 hPa, 150 hPa, 200 hPa, 250 hPa, 300 hPa, 400 hPa, 500 hPa, 600 hPa, 700 hPa, 850 hPa, 925 hPa and 1000 hPa) and 4 surface variables. 13 out of 37 pressure levels and surface-level variables, including geopotential, specific humidity, temperature, and wind speed	ECMWF (ERA5)	40 years	39 years	1 year
Lam et al. (2023) (GraphCast)	ECMWF (ERA5)	Hundreds of weather variables (temperature, wind components, mean sea-level pressure, geopotential, total precipitation, specific humidity, and vertical wind speed in 37 pressure levels	ECMWF (ERA5) and High RESolution forecast (HRES)	43 years	39 years	4 years
Chen et al. (2023a) (FuXi)	ECMWF (ERA5)	Meteorological parameters including 5 upper-air atmospheric variables at 13 pressure levels (50, 100, 150, 200, 250, 300, 400, 500, 600, 700, 850, 925, and 1000 hPa), and 5 surface variables. The 5 upper-air atmospheric variables are geopotential, temperature, wind components, and relative humidity. Additionally, 5 surface variables are temperature, wind components, mean sea-level pressure, and 6-hourly total precipitation,	ECMWF (ERA5)	40 years	39 years	1 year
Chen et al. (2023b) (FengWu)	ECMWF (ERA5)	Various weather variables, including upper-air variables (geopotential, specific humidity, temperature, and wind speed) at different pressure levels, as well as surface variables (temperature, wind speed, and mean sea level pressure).	ECMWF (ERA5)	40 years	39 years	1 year
Hu et al. (2023)	WB data (a preprocessed subset of ERA5)	Geopotential, temperature, relative humidity, longitude-direction wind, and latitude-direction wind at all 13 vertical layers, four single-layer fields (temperature, wind, and total precipitation), and two constant fields (land-sea mask and orography)	WB data	40 years	38 years	2 years
Lagomarsino-Oneto et al. (2023)	Meteorological stations	Wind speed and wind direction	Meteorological station	Time span between 4 and 7 years	–	–

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Table 5 (continued)

Study	Training and validation data	Parameter	Testing data	Data time span	Training and validation time span	Testing time span
Nguyen et al. (2023a) (ClimateLearn)	ECMWF (ERA5) and CMIP6	Different climate variables at 37 different pressure levels	ECMWF-Integrated Forecasting System (IFS)	39 years	37 years	2 years
Nguyen et al. (2023b) (ClimaX)	ECMWF (ERA5) and CMIP6	Different climate variables at 37 different pressure levels	ECMWF-IFS	40 years	38 years	2 years
Zenkner and Navarro-Martinez (2023)	Meteorological stations	Air temperature, relative humidity, and wind speed	Meteorological station	5 years	4.25 years	0.75 years
Hua et al. (2024)	WB data (a preprocessed subset of ERA5)	Geopotential at 500 hPa and temperature at 850 hPa	WB data	39 years	37 years	2 years
Gao et al. (2024) (PreDiff)	Storm Event ImageRy (SEVIR)	Meteorological parameters such as cloud cover, atmospheric conditions, and precipitation	SEVIR	–	–	–
Lang et al. (2024) (AIFS)	ECMWF (ERA5)	Various weather variables, including geopotential, specific humidity, temperature, vertical velocity, and wind speed at 13 pressure levels, as well as surface variables such as temperature, wind speed, and mean sea level pressure	IFS and ERA5	44 years	41 years	3 years
Price et al. (2025) (GenCast)	ECMWF (ERA5)	Various weather variables, including geopotential, specific humidity, temperature, vertical velocity, and wind speed at 13 pressure levels, as well as surface variables such as temperature, wind speed, and mean sea level pressure	ECMWF ensemble forecast	40 years	39 years	1 year

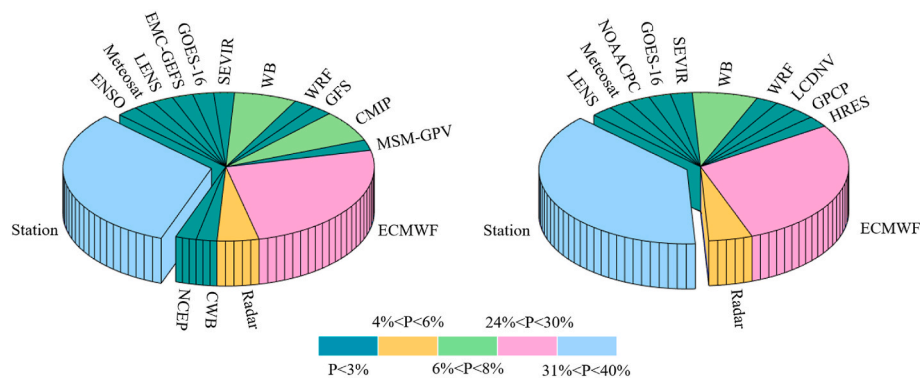


Fig. 4. Contribution of the different datasets in AI-based models (Left: training and validation, Right: testing). 'P' value represents the contribution percent of each dataset.

where the atmosphere's inherent complexity and variability necessitate sophisticated methods to evaluate forecast confidence (Qian et al., 2016). Historically, NWP has employed statistical and probabilistic tools to manage these uncertainties. A prominent example is ensemble forecasting, which involves running multiple instances of models with varying inputs or parameters to generate a range of possible outcomes. Ensemble forecasting captures the uncertainties inherent in weather predictions by providing a spectrum of potential results rather than a single deterministic forecast. This approach is crucial for risk assessment, decision-making, and improving the forecasting of extreme weather events. However, raw ensemble forecasts often suffer from biases and calibration errors that reduce their reliability (Buizza, 2018). In the field of DL, methods such as dropout (originally introduced to prevent overfitting) have been reinterpreted as Bayesian techniques that approximate uncertainty by producing variable outputs through repeated stochastic forward passes. Moreover, architectures explicitly designed for uncertainty modeling, including probabilistic NNs and deep ensembles, provide enhanced capabilities for capturing prediction confidence (Tang et al., 2022). The integration of DL models with NWP systems offers a more comprehensive understanding of uncertainty throughout both the simulation and prediction stages (Cheng et al., 2023). CNNs, GANs, and Temporal NNs (TNNs) have all shown promise

in this domain. CNNs have demonstrated success in refining spatial representations, thereby improving accuracy in daily precipitation forecasting tasks (Wang et al., 2024). GANs are increasingly employed to produce fine-grained, probabilistic weather forecasts by learning from the distribution of ensemble errors. This enables the generation of synthetic ensemble members that better reflect observed datasets (Glawion et al., 2023). Meanwhile, TNNs have emerged as valuable tools for enhancing ensemble forecast post-processing. Their attention-based architectures allow models to capture complex temporal dependencies and sequential structures, a critical capability for predicting rare and extreme weather events (Mandal, 2024). Chen et al. introduced FuXi Subseasonal-to-Seasonal (FuXi-S2S) (Chen et al., 2024), a model inspired by VAE architecture, capable of providing global daily mean forecasts up to 42 days using ensemble forecasting. Trained on ECMWF ERA5 data, FuXi-S2S exhibited superior skill compared to existing models, particularly improving predictions of precipitation and the Madden-Julian Oscillation (MJO). Building on this framework, Zhong et al. (2024b) developed FuXi-ENS, an advanced machine learning system designed for 6-hourly global ensemble forecasts up to 15 days. This model addresses limitations in traditional ensemble prediction systems by operating at a high spatial resolution of 0.25° , incorporating a comprehensive set of atmospheric and surface variables across

Table 6
Specifications of current AI-based weather forecasting models.

Study	Predictand parameters	Extreme events	Spatial scale	Lead time	Temporal resolution
Kuligowski and Barros (1998)	Precipitation	Heavy precipitation	Four specific locations in the middle Atlantic region of the United States	Short-term (24-h)	Hourly
Radhika and Shashi (2009)	Daily maximum temperature	–	One specific location in England	Short to medium-term (2–10-day)	Daily
Wei (2012)	Precipitation	Tropical cyclone events	One specific location in Taiwan	Short-term (up to 72 h)	Hourly
Sharma et al. (2014)	Rain, Snow, and Fog	–	One specific station	Short-term (daily)	Daily
Liu et al. (2015)	Temperature and wind speed	–	One specific location in Hong Kong	Short-term (daily)	Hourly
Mislan et al. (2015)	Precipitation	–	One specific station in Indonesia	Long-term (Monthly)	Monthly
Salman et al. (2015)	Precipitation	–	One specific location in Indonesia	–	–
Shi et al. (2015)	Precipitation	–	Hong Kong	Short-term (6 h)	Hourly
Hernández et al. (2016)	Accumulated daily precipitation	–	One specific station in Colombia	Short-term (daily)	Daily
Kashiwao et al. (2017)	Precipitation	Heavy precipitation	Multiple stations in Japan	Short-term (daily)	Daily
Qiu et al. (2017)	Precipitation	–	One specific station in Colombia and multiple stations in China	Short-term (3 h)	Hourly
Zhang et al. (2017)	Precipitation	–	One specific station in China	Short-term (daily)	Daily
Karevan and Suykens (2018)	Minimum and maximum temperature	–	Multiple stations in 5 cities including Brussels, Antwerp, Liege, Amsterdam and Eindhoven	Short to medium-term (1 to 6-day)	Daily
Larraondo et al. (2019)	Total precipitation	–	A part of Europe	Short-term (6-h)	6-h
Chattopadhyay et al. (2020)	Temperature	Cold and heat waves	North American continent	Short to medium-term (1 to 5-day)	Daily
Peng et al. (2020)	Temperature	–	East Asia	Short to long-term (1–35-day)	Daily
Roy (2020)	Temperature	–	One specific station in United States	Short to medium-term (1 and 10-day)	Daily
Sønderby et al. (2020) (MetNet)	Precipitation	–	Continental United States	Short-term (up to 8 h)	2-min
Weyn et al. (2020)	Several basic atmospheric variables including: geopotential at 500 hPa and at 1000 hPa, 300- to 700-hPa geopotential thickness, and 2-m temperature	–	Global	Short to medium-term (up to 7 days)	6-h
Hanoon et al. (2021)	Temperature and relative humidity	–	One specific station in Malaysia	Short and long-term (daily and monthly)	Daily and monthly
Hewage et al. (2021)	Surface temperature, surface pressure, wind direction, specific humidity, convective rain, non-convective rain, snow water equivalent, soil temperature, and soil moisture	–	One specific location	Short-term (3, 6, 9, 12, 24 and 48-h)	3-h
Kwok and Qi (2021)	Temperature, convective precipitation rate intensity, cloud mask, and probability of tropopause folding	–	Different regions in Europe and Africa	Short-term (8-h)	15-min
Lin et al. (2022)	Temperature, cloud cover, humidity, and surface wind components	–	Global	Short-term (1–12-h)	Hourly
Keisler (2022)	Temperature, geopotential, specific humidity, eastward wind component, northward wind component, and vertical wind component	–	Global	Short-term (6-h)	6-h
Pan et al. (2022)	Monthly precipitation and temperature	–	Global	Long-term (3–8 months)	Monthly
Pathak et al. (2022) (FourCastNet)	20 factors, including one parameter present at four different levels, two parameters at three levels, two others at two levels, five parameters at the surface level, and one parameter at the integrated level.	Hurricanes and atmospheric rivers	Global	Short to medium-term (up to 10 days)	6-h
Yin et al. (2022)	Precipitation	Extreme precipitation	Japan	Short-term (39 h)	Hourly
Ang (2023)	Precipitation	–	China	Short-term	3-h
Bi et al. (2023) (Pangu-Weather)	69 factors, including 5 upper-air variables at 13 pressure levels (50 hPa, 100 hPa, 150 hPa, 200 hPa, 250 hPa, 300 hPa, 400 hPa, 500 hPa, 600 hPa, 700 hPa, 850 hPa, 925 hPa and 1000 hPa) and 4 surface variables.	Tropical cyclones	Global	Short to medium-term (up to 7 days)	6-h
Lam et al. (2023) (GraphCast)	227 factors, including 6 upper-air variables at 37 pressure levels and 5 surface variables. The variables include temperature, wind components,	Tropical cyclones, atmospheric rivers, and extreme temperatures	Global	Short to medium-term (up to 10 days)	6-h

(continued on next page)

Table 6 (continued)

Study	Predictand parameters	Extreme events	Spatial scale	Lead time	Temporal resolution
Chen et al. (2023a) (FuXi)	mean sea-level pressure, geopotential height, total precipitation, specific humidity, and vertical wind speed.	–	Global	Short to long-term (15 days forecast horizon)	6-h
Chen et al. (2023b) (FengWu)	Geopotential, relative humidity, temperature, wind components, total precipitation, and mean sea-level pressure	–	Global	Short to medium-term (10 days forecast horizon)	6-h
Hu et al. (2023)	Various atmosphere parameters, including surface and atmosphere variables at different pressure levels, such as geopotential, humidity, wind components, and temperature	–	Global	Short to long-term (up to 14 days)	6-h
Lagomarsino-Oneto et al. (2023)	Number of meteorological parameters including geopotential, temperature at 850 hPa, surface temperature, and total precipitation	–	32 stations in Italy	Short-term (1, 3, 6, 12, 18, and 24-h)	Hourly
Nguyen et al. (2023a) (ClimateLearn)	Wind speed	–	Global	Short to medium-term (6-h, 1, 3, 5, and 10-day)	6-h
Nguyen et al. (2023b) (ClimaX)	High number of variables including geopotential, temperature, wind components, relative humidity, and specific humidity at different pressure levels	Extreme weather	Global	Short to long-term (6 h, 1, 3, 5, 7, 10, 14, and 30 days)	6-h
Zenkner and Navarro-Martinez (2023)	Geopotential, temperature, zonal and meridional wind, relative humidity, and specific humidity at different pressure levels, 2-m temperature, wind components	–	Global and regional (North America)	Short to long-term (up to 72 h)	Hourly
Hua et al. (2024)	temperature, relative humidity, and wind speed	–	Two specific stations in England	Short-term (up to 14 days)	6-h
Gao et al. (2024) (PreDiff)	Geopotential at 500 hPa and the temperature at 850 hPa	–	Global	Short-term (1 h)	10-min
Lang et al. (2024) (AIFS)	Precipitation	Rainstorms and droughts	Specific location	Short to medium-term (up to 10 days)	3-h- After the sixth day, it will become 6-h
Price et al. (2025) (GenCast)	Different weather variables (temperature, wind components, mean sea-level pressure, geopotential, total precipitation, specific humidity, and vertical wind speed at 13 pressure levels and a few surface variables	–	Global	Short to medium-term (up to 15 days)	12-h
	Many factors such as temperature, wind components, mean sea-level pressure, geopotential, total precipitation, specific humidity, and vertical wind speed at 13 pressure levels and a few surface variables	Tropical cyclone tracks and wind power production.	Global		

multiple pressure levels. By employing a VAE-based loss function that combines Continuous Ranked Probability Score (CRPS) and KL divergence, FuXi-ENS improves the representation of flow-dependent perturbations and outperformed ECMWF in 98.1 % of CRPS comparisons, demonstrating its potential to significantly enhance ensemble weather forecasting. Despite these advances, significant challenges remain in applying DL-based methods for uncertainty approximation. These models often require careful hyperparameter tuning and are prone to overfitting, which can undermine their generalization ability. Additionally, interpreting uncertainty estimates produced by deep models is not always straightforward. Given the complexity of atmospheric dynamics, there is a risk that DL models may fail to fully represent the underlying physical processes, potentially resulting in misleading uncertainty assessments.

In meteorology, an extreme event refers to an unusual occurrence within the statistical distribution of any given weather parameter (Herring et al., 2018). Such events include extremes in temperature, excessive precipitation, intense storms, or strong winds. Some studies summarized in Table 6 tested their models under extreme weather conditions. Among these, FuXi-Extreme (Zhong et al., 2024a) stands out as a significant advancement. This model enhances the original FuXi model introduced by Chen et al. (2023a) by incorporating a denoising diffusion probabilistic module that effectively restores finer-scale details of surface meteorological variables. As a result, FuXi-Extreme outperforms both the ECMWF-HRES and the original FuXi models in forecasting extreme weather events, including the tracking and prediction of tropical cyclone intensity. Complementing the FuXi models, Price et al. developed GenCast (Price et al., 2025), a probabilistic weather model

based on diffusion techniques. Trained on ERA5 data, GenCast outperformed the ECMWF ensemble in more than 97 % of cases. It delivers high-resolution 15-day global forecasts and significantly improves predictions of extreme weather and wind power generation.

5. Evaluation metrics and forecasting results

Current AI-based weather forecasting models utilize a diverse range of metrics to evaluate their performance and efficacy in weather forecasting. These metrics offer insights into various aspects of model performance, including accuracy, precision, reliability, and skill. The list of metrics used in performance evaluation and their usage frequency across different studies is shown in Table 7. It reflects the relative popularity and adoption of various statistical parameters used to assess model performance. Metrics like RMSE/NRMSE (62.5 %) and ACC (25 %) were among the most frequently used, suggesting a strong emphasis on error magnitude and anomaly detection.

Table 8 presents the summary and the most important statistical results obtained from current AI-based models. It is important to acknowledge that error metrics such as RMSE, MAE, and MSE, which are not normalized, may not be directly comparable across various climatological regions or different seasons within the same region. For example, an RMSE of 1 mm/day in precipitation forecasting may have different implications in a region with a mean daily precipitation of 5 mm compared to a region with a mean daily precipitation of 20 mm. Therefore, while these metrics provide valuable insights into the performance of AI-based weather forecasting models, caution should be exercised when interpreting and comparing results across different

Table 7
Evaluations metrics and their specifications.

Metric	Specification	Usage frequency
Root Mean Square Error (RMSE)	Indicating the average magnitude of errors between predicted and observed values	62.5 %
Normalized Root Mean Square Error (NRMSE)	A normalized perspective, facilitating comparisons across models with different units and scales	
Mean Squared Error (MSE)	Alternative views of error magnitude, with emphasizing larger errors	17.5 %
Normalized Mean Square Error (NMSE)	A scale-independent evaluation metric that measures the MSE relative to the variance of the observed data	
Mean Absolute Error (MAE)	Alternative views of error magnitude, with focusing on the average magnitude of errors	22.5 %
Bias	A measure of systematic overestimation or underestimation tendencies within the model's predictions	10 %
Estimated Threat Score (ETS)	Evaluates forecast skill relative to a reference forecast, incorporating hits and false alarms	2.5 %
Precision and accuracy	Highlighting the proportion of correct positive predictions and overall correct predictions, respectively	20 %
R ² and the Coefficient of Correlation (R)	Assess the strength and direction of the relationship between two variables	12.5 %
Critical Success Index (CSI), Probability of DEtection (PODE)	Assess the ability of the system to correctly identify events of interest, such as the occurrence of weather phenomena or the presence of objects in a surveillance system	12.5 %
Hit Rate of Precipitation (HRP)	Addresses the accuracy of precipitation event predictions	2.5 %
CRPS	Assesses probabilistic forecast accuracy by comparing cumulative distribution functions	17.5 %
Anomaly Correlation Coefficient (ACC)	Examines the correlation between predicted and observed anomalies, offering insights into the model's ability to capture unusual or anomalous weather patterns	25 %

contexts. This underscores the critical need for standardized common benchmarks to enable fair and meaningful comparisons of AI-based weather forecasting models.

In a joint effort by researchers from Google and the ECMWF, WB2 was developed as an open, standardized benchmarking framework using the same dataset and conditions for evaluating medium-range weather forecasts generated by data-driven and AI-based models (Rasp et al., 2024). WB2 aligns with established verification protocols from operational centers such as ECMWF and the World Meteorological Organization (WMO), providing a robust and adaptable evaluation environment. Rather than relying on a single metric or fixed leaderboard, WB2 integrates a comprehensive set of headline metrics and diagnostic tools designed to capture the multidimensional nature of forecast skill. A key innovation of WB2 is its explicit support for probabilistic forecasting, reflecting the intrinsic uncertainty inherent in atmospheric systems. It is recommended to adopt WB2 as a standardized evaluation framework to enable consistent, fair, and meaningful comparisons of AI-based weather forecasting models.

5.1. Overfitting

Overfitting is a significant challenge in AI-based weather forecasting, occurring when a model performs exceptionally well on training data but fails to generalize effectively to new, unseen data. This results in reduced prediction accuracy and undermines the model's reliability in

Table 8
Statistical results of the AI-based weather forecasting models.

Study	Result
Kuligowski and Barros (1998)	The NN method have lower RMSE's than the regression forecasts for observed precipitation amounts of 0.20 in. (5.1 mm) or greater in 6-h prediction. For a threshold of 1.00 in. (25.4 mm), the reduction in rmse over regression exceeds 10 %.
Radhika and Shashi (2009)	The MSE in the case of MLP varies from 8.07 to 10.2 °C for daily maximum temperature, whereas it is in the range of 7.07–7.56 °C in the case of SVM.
Wei (2012)	Results show that better attribute combinations for typhoon climatologic characteristics and typhoon precipitation predictions occurred at 0-h lag time with modified RMSE values of 0.288, 0.257, and 0.296 mm/h in GSVM, WSVM, and the regression, respectively. Moreover, WSVM having average bias and ETS values close to 1.0 gave better predictions than did the GSVM and regression models.
Sharma et al. (2014)	Fog: Precision achieved for values of K = 25, 30,35,40,45 and 50 using KNN method. The achieved Precision remains 0 % for different values of K. Rain: Precision achieved for values of K = 25, 30,35,40,45 and 50. The achieved precision remains above 67 % for different values of K. Snow: Precision achieved for values of K = 25, 30,35,40,45 and 50. The achieved precision shows that the maximum amount is more than 30 % for k = 40.
Liu et al. (2015)	The results show the NRMSE of 2.179e-2 and 8.117e-3 °C for classical SVR and NN with SVR in top layer, respectively in temperature prediction. The results show the NRMSE of 0.3721 and 0.2522 m/s for classical SVR and NN with SVR in top layer, respectively in wind speed prediction.
Mislan et al. (2015)	The experimental results show the MSE of 0.00096341 m for precipitation prediction.
Salman et al. (2015)	The performance of RNN for precipitation prediction shows the R ² value reached a maximum of 84.8 %.
Shi et al. (2015)	The results of ConvLSTM method show the MSE of 1.42, CSI of 0.577, PODE of 0.660, and correlation of 0.908
Hernández et al. (2016)	The results show the ability of integration of autoencoder and MLP method with RMSE of 6.33 mm
Kashiwao et al. (2017)	The results show the ability of the MLP method with a Hit Rate of Precipitation (HRP) of 64.9 mm in one of the specific locations of the study area, in comparison with the RBFN method with an HRP of 45.2.
Qiu et al. (2017)	The results show the ability of the MT-CNN method with a MSE of 11.253 mm and correlation of 0.521.
Zhang et al. (2017)	The results show the ability of the DBNPF method with a MSE of 2.257 cm and MAE of 1.632.
Karevan and Suykens (2018)	The results show the ability of the spatiotemporal SS-LSTM method with a MSE of 5.40 °C in 6-day prediction, in comparison with the stacked LSTM method with an MSE of 22.06 °C in maximum temperature prediction.
Larraondo et al. (2019)	The results show that the lowest MAE after training steps is observed in U-Net, with an MAE of 0.2386 mm.
Chattopadhyay et al. (2020)	Using both surface temperature and geopotential at 500 hPa, accuracies (recalls) with CapsNets increase to ~ 80 % (88 %). In both cases, CapsNets outperform simpler techniques such as CNNs and regression.
Peng et al. (2020)	The MLP is slightly better than NGBoost, with a mean RMSE of 3.202 °C. In terms of convenience and reliability, the MLP is more appropriate and superior for extended-range temperature forecasts. However, NGBoost is more flexible and is designed to develop probabilistic forecasts. From the perspective of probabilistic forecasts, NGBoost is better than the NN in terms of average CRPS, with a value of 1.81.

(continued on next page)

Table 8 (continued)

Study	Result
Roy (2020)	Integration of CNN and LSTM shows the best performance with an RMSE of about 0.5 F for 1-day prediction and about 0.4 F for 10-day prediction.
Sønderby et al. (2020) (MetNet)	A kind of Integration of CNN and LSTM, called MetNet, shows the best performance up to 400–480 min with maximum f1 score of about 0.3. Additionally, the paper mentions that MetNet is the first ML model to outperform NWP on a structured prediction task at such a range and resolution.
Weyn et al. (2020)	The RMSE geopotential at 500 hPa shows that the model significantly outperforms persistence at all lead times and climatology up to more than 7 days. The RMSE on the sixth day of forecasting is about 700 m ² /s ² , while the RMSE for temperature on the same day is about 2.4 K.
Hanoon et al. (2021)	Results showed that MLP performs well among the others in predicting daily temperature and relative humidity with R of 0.7132 and 0.633, respectively. However, in monthly prediction temperature also MLP model provided closer standards deviation to actual value and can be used to predict monthly temperature with R 0.8462. Whereas in prediction monthly relative humidity, the RBFN model's efficiency was higher than other models with R of 0.7113.
Hewage et al. (2021)	The MSE for MIMO-LSTM in wind speed for the U and V directions in the 3-h prediction was 5.32 and 4.62 m/s, respectively—both lower than the results of the WRF model. For very long-term forecasting, the WRF model performed better than the ML learning model.
Kwok and Qi (2021)	The test leaderboard scores for the core challenge ranged from 0.4977 to 0.8822, while for the transfer learning challenge, they ranged from 0.4878 to 0.6111.
Lin et al. (2022)	In the 12-h temperature prediction, the CLCRN method shows the RMSE and MAE to be about 1.16 K and 1.88 K lower, respectively, than those of other methods. Further analysis reveals two existing problems with the method: the over-smoothness of the learned local patterns and the instability of the training process.
Keisler (2022)	The comparison with LCDNV data reveals that the RMSE for temperature at the 850 hPa level in the initial forecast time is approximately 0.8 K, which is close to that of GFS but higher than that of ERA5.
Pan et al. (2022)	Results indicate that the proposed CNN-based method demonstrates competitive performance compared to dynamical forecasts. The ACC of the proposed method is 0.4.
Pathak et al. (2022) (FourCastNet)	The model shows the ACC and RMSE of geopotential at 500 hPa for 24-h prediction (7 days forecast horizon) is 0.762 and 595, respectively.
Yin et al. (2022)	The SVM-based method shows the ACC and RMSE of geopotential at 500 hPa for 24-h prediction (7 days forecast horizon) is 0.762 and 595, respectively.
Ang (2023)	U-Net achieves an accuracy of 96.89 % in predicting precipitation based on the 6-month test results. Transformer achieves an accuracy of 94.13 % for the same task.
Bi et al. (2023) (Pangu-Weather)	The ACC and RMSE of geopotential at 500 hPa for 24-h prediction (7 days forecast horizon) is 0.872 and 510, respectively (day 7).
Lam et al. (2023) (GraphCast)	The ACC and RMSE of geopotential at 500 hPa for 24-h prediction (10 days forecast horizon) is 0.825 and 460, respectively (day 7) better than HRES.
Chen et al. (2023a) (FuXi)	The CRPS for the first day of forecasting temperature at the 850 hPa level is lower than 0.5. However, by day 15 of the forecast, the CRPS exceeds 2.
Chen et al. (2023b) (FengWu)	The proposed model reduces the RMSE of the 10-day lead global geopotential at 500 hPa prediction from 733 to 651 m ² /s ² compared to the model proposed by Lam et al. (2023).
Hu et al. (2023)	The performance of all the evaluated four variables achieves remarkable improvements for both 5 and 14 days forecasts. Specifically, the RMSE scores

Table 8 (continued)

Study	Result
	decrease from 431/972 (SwinRNN) to 401/795 (SwinVRNN) for geopotential at 500 hPa, 2.20/4.25 to 2.07/3.47 for temperature at the 850 hPa level, 1.75/3.28 to 1.64/2.70 for 2-m temperature, and 2.17/2.40 to 2.16/2.30 for 6-hourly total precipitation.
Lagomarsino-Oneto et al. (2023)	The comparison with the persistence model shows the RMSE of about 0.95 for the first hour of the prediction, lower than competing models.
Nguyen et al. (2023a) (ClimateLearn)	The ACC and RMSE of ResNet in prediction of geopotential at 500 hPa is 0.7 (higher than U-Net and ViT, lower than IFS) and 740 m ² /s ² (lower than U-Net and ViT, higher than IFS), respectively (day 7). The ACC and RMSE of ResNet in prediction of temperature at the 850 hPa is 0.775 (higher than U-Net and ViT, lower than IFS) and 3 K (lower than U-Net and ViT, higher than IFS), respectively (day 7).
Nguyen et al. (2023b) (ClimaX)	The ACC and RMSE of the model in prediction of geopotential at 500 hPa is 0.8 (higher than U-Net and ResNet, lower than IFS) and 620 m ² /s ² (lower than U-Net and ResNet, close to IFS), respectively (day 7). The ACC and RMSE of the model in prediction of temperature at the 850 hPa is 0.82 (higher than U-Net and ResNet, lower than IFS) and 2.8 K (lower than U-Net and ResNet, higher than IFS), respectively (day 7).
Zenkner and Navarro-Martinez (2023)	The model shows the RMSE of 2.26 °C and 14 %, for air temperature and relative humidity predictions, respectively.
Hua et al. (2024)	The RMSE for geopotential at 500 hPa ranges from less than 100 m ² s ⁻² to over 1100 m ² s ⁻² . Similarly, RMSE for temperature at 850 hPa spans from less than 1 K to under 5 K.
Gao et al. (2024) (PreDiff)	The CSI of precipitation at a pooling scale of 16 × 16 was about 0.6244.
Lang et al. (2024) (AIFS)	The ACC for geopotential at 500 hPa is greater than 0.90 on the fifth day and exceeds 0.50 on the tenth day.
Price et al. (2025) (GenCast)	The greatest improvements occur at shorter forecast periods of approximately 3–5 days, particularly for surface variables and for temperature and specific humidity at higher pressure levels, where CRPS skill scores are improved by 10 %–30 % compared to ECMWF ensemble forecast.

real-world applications. In the context of weather forecasting, overfitting is often exacerbated by the vast and complex nature of the input data, which spans the entire Earth. Models, particularly those employing global attention mechanisms, may inadvertently learn arbitrary or spurious relationships instead of capturing meaningful meteorological patterns. To address these challenges, researchers have explored strategies at both the architectural and training levels. From an architectural perspective, incorporating locality bias, which leverages the spatial and temporal coherence inherent in weather systems, has proven effective. Models based on CNNs (Cheon et al., 2024), GNNs as used in GraphCast and AIFS (Lam et al., 2023; Lang et al., 2024), and local attention mechanisms as implemented in Pangu-Weather, FengWu, and FuXi-2.0 (Bi et al., 2023; Chen et al., 2023b; Zhong et al., 2024c) help constrain the model to focus on relevant local interactions. This reduces the risk of overfitting to irrelevant global noise. Additionally, pretraining approaches designed to embed locality bias into the model have shown promise, although this strategy remains relatively underexplored in the literature. Complementing these architectural strategies are several well-established training techniques aimed at mitigating overfitting. Layer-wise unsupervised pretraining enables more effective weight initialization, especially when labeled data is scarce. L2-norm regularization penalizes large weights to discourage overly complex models. Dropout regularization randomly deactivates neurons during training to prevent co-adaptation and improve robustness. Early stopping monitors validation performance and halts training when

overfitting begins to occur. More recent methods such as scheduled droppath probabilistically drop entire layers during training, exposing the model to diverse subnetworks and enhancing generalization (Table 9). Despite the importance of these techniques, relatively few studies explicitly report their use. In many cases, the problem of overfitting is acknowledged only briefly or omitted altogether. This highlights a gap in the current literature and points to a need for more systematic evaluation and reporting of overfitting mitigation strategies in AI-based weather forecasting models.

5.2. Data assimilation

The modern forecasting system includes three components: the observation component, the assimilation system (diagnostic or analysis component), and the numerical model (prognostic component) (Pu and Kalnay, 2019). The observation phase collects atmospheric data, which is then combined with previous model outputs during data assimilation to establish accurate initial conditions for generating forecasts. Most modern AI-based weather forecasting models utilize the ECMWF ERA5 dataset, which relies on the Four-Dimensional Variational (4D-Var) assimilation system. However, this conventional approach has intrinsic limitations. It heavily depends on fixed physical models and pre-determined error statistics, assumes the system behaves in a linear or

Table 9
Overfitting solutions used in AI-based weather forecasting models.

Overfitting solution	Specifications	Used by
Layer-wise Unsupervised Pre-training	Involves training each layer of a network individually using unsupervised learning techniques such as autoencoders. By initializing the model with pre-trained weights, it can help the model to converge to a better solution, especially when labeled training data is scarce.	• Liu et al. (2015)
L2-norm Regularization	Involves adding a penalty term to the loss function proportional to the square of the magnitude of the weights. This encourages the model to learn smaller weight values, preventing it from becoming overly reliant on a few input features and reducing the risk of overfitting.	• Karevan and Suykens (2018)
Dropout Regularization	Randomly deactivates a proportion of neurons in a NN during training. By temporarily removing connections between neurons, dropout prevents the model from relying too heavily on specific features or co-adapting to the training data, thus improving its generalization performance.	• Chattopadhyay et al. (2020) • Kwok and Qi (2021) • Zenkner and Navarro-Martinez, 2023
Early Stopping	Involves monitoring the model's performance on a separate validation dataset during training and halting the training process when the validation error starts to increase, indicating that the model is beginning to overfit the training data.	• Pan et al. (2022)
Scheduled Droppath	Instead of randomly dropping individual neurons, entire layers are dropped during training with a certain probability. This approach encourages the model to learn more robust representations by exposing it to different subnetworks throughout the training process.	• Chen et al. (2023a)

near-linear manner, and faces challenges incorporating unconventional, high-resolution observations without losing valuable information through resampling or interpolation.

Errors arising during data assimilation generally fall into two categories: inaccuracies in initial conditions and deficiencies in the forecasting models themselves. Traditional assimilation techniques, such as the Kalman filter, typically treat these errors as additive, uncorrelated (white) noise with zero mean. Although these assumptions simplify the error modeling process, they often do not hold true in reality. For instance, operational forecasting centers like ECMWF have relaxed some of these assumptions, using multiplicative error models to better capture error characteristics (Buizza et al., 1999). The zero-mean error assumption is especially limiting because it overlooks systematic biases in models that can vary across space, time, and weather conditions (Dee, 2005). Many NWP systems do not explicitly model these biases, opting instead for indirect corrections like debiasing residuals or inflating background error covariances. Techniques such as variational bias correction (Auligné et al., 2007) and background error inflation (Houtekamer and Zhang, 2016) address some of these issues. A more comprehensive method is the Weak-Constraint 4D-Var (WC-4D-Var), which extends the assimilation framework by explicitly accounting for model error tendencies (Tremolet, 2006). ECMWF's Integrated Forecasting System (IFS) Cycle 47R1 recently demonstrated operational improvements, particularly in reducing biases in the stratosphere, using this approach (Laloyaux et al., 2020).

In recent years, AI-based data assimilation methods have emerged that enhance traditional approaches by employing ML and DL to better identify and correct model errors and biases. These hybrid methods integrate physics-based models with ML to boost both precision and adaptability. ML is often utilized through surrogate modeling, where NNs approximate complex components of the system, lowering computational demands while preserving critical dynamics. Unlike conventional data assimilation, which depends on physical assumptions and statistical models, AI-based methods learn directly from historical data and simulation outputs. Examples include RNNs (Park, 2010) and ResNets (Brajard et al., 2020), which are designed to capture the underlying system dynamics or response functions (Fablet et al., 2018; Long et al., 2018; Bocquet et al., 2020). A frequent challenge with AI-based assimilation methods is the unrealistic assumption of complete observability, whereas actual observations are often sparse and noisy. To overcome this, Brajard et al. (2020) proposed an approach that alternates between the model training and assimilation updates, allowing both the surrogate model and state estimates to be refined simultaneously. Bocquet et al. (2020) further developed this idea within a Bayesian framework to better handle uncertainty. Additionally, CNNs applied across multiple dimensions have been used to process image-based observational data, enhancing feature extraction for assimilation tasks such as estimating tropical cyclone intensity (Lee et al., 2019).

Despite these promising advances, AI-based data assimilation faces several challenges. These methods can be computationally demanding, sensitive to initial conditions, and prone to instability. A common strategy to address these issues is to combine DL components with existing physics-based models rather than completely replacing them (Pathak et al., 2018; Watson, 2019; Casas et al., 2020; Bonavita and Laloyaux, 2020). Nevertheless, problems persist, including the need for large datasets with labeled training examples and the difficulty of reconciling differences in spatial and temporal resolutions between model data and observations.

5.3. Data-driven models

The main challenge in AI-based weather forecasting lies in developing truly data-driven models that can operate independently of traditional NWP systems. Conventional forecasting relies heavily on a complex pipeline involving observation gathering, data assimilation,

and numerical modeling to produce forecasts. This system integrates current and past atmospheric data with physics-based numerical models, treating weather forecasting as an initial value problem that depends on these numerical models for initialization and prediction. Current AI-based models, while promising, still depend on this numerical modeling component, especially at prediction time, which limits their ability to fully replace conventional systems.

Data-driven models Aardvark have been developed to overcome these limitations by replacing the entire NWP pipeline with a single data-driven system. Data-driven models directly learn to map raw observational data to global and local weather forecasts without relying on any numerical model inputs at deployment. This approach dramatically reduces computational costs and complexity, enabling much faster and more flexible forecasting while maintaining or even improving accuracy compared to operational NWP systems.

Aardvark Weather is a groundbreaking DL system that redefines weather forecasting by eliminating reliance on traditional physics-based models at deployment. Instead, it learns a direct mapping from observational data to forecasts. The system comprises three integrated modules: an encoder that estimates the initial atmospheric state from heterogeneous data both on and off the grid, a processor that autoregressively generates global forecasts, and a decoder that converts these into local predictions for weather stations. At its core, Aardvark uses a vision transformer for encoding and processing, paired with a lightweight convolutional decoder for local outputs. Aardvark produces global forecasts for variables like temperature, wind, humidity, and pressure on a 1.5° grid, and local forecasts for 2-m temperature and 10-m wind speed, achieving skillful performance up to ten days ahead. Training mirrors deployment: the encoder is pretrained on raw observations using high-resolution ERA5 reanalysis data as targets, the processor is trained on ERA5 inputs and outputs, and the decoder is trained using model outputs against raw station data. Final end-to-end fine-tuning optimizes for specific regions or variables. Remarkably, Aardvark uses only about 8 % of the observational data required by traditional numerical weather prediction systems, yet matches or surpasses them in accuracy and lead time. It ingests diverse data sources, including in situ measurements from stations, ships, and radiosondes, as well as satellite-based remote sensing like scatterometers and infrared sounders. These inputs, structured as gridded and non-gridded formats, are transformed into an initial atmospheric state via the encoder. ERA5 reanalysis plays a key role in training, helping the model learn and correct observational biases without explicit bias correction. One of Aardvark's most transformative features is its speed and efficiency. It delivers forecasts in seconds using only a few Graphics Processing Units (GPUs), compared to the thousands of supercomputer node hours required by conventional NWP systems. This makes rapid, scalable, and cost-effective deployment possible, including in regions with limited computational resources (Allen et al., 2025).

One of the most important limitations of these models is underperforming in predicting extreme weather events because such conditions are underrepresented in their training data, leading to potential underestimation of risks. Finally, the evaluation of these AI models is often incomplete and lacks standardized benchmarks, underscoring the need for improved methods to enhance their credibility and robustness.

6. SOTA AI-based weather forecasting models

Based on advancements in method, spatial scale, and performance, the focus of this section is analyzing seven specific models, including FengWu (Chen et al., 2023b), ClimaX (Nguyen et al., 2023b), Pangu-Weather (Bi et al., 2023), FourCastNet (Pathak et al., 2022), GraphCast (Lam et al., 2023), GenCast (Price et al., 2025), and AIFS (Lang et al., 2024).

6.1. FengWu

FengWu represents another significant advancement in weather forecasting (Chen et al., 2023b). It uses a multi-modal and multi-task approach based on a combination of model-specific encoders and decoders alongside a cross-modal fusion transformer architecture. This design enables FengWu to effectively capture and analyze diverse data modalities. One of the key features distinguishing FengWu is its innovative treatment of uncertainty. By integrating an uncertainty loss mechanism, FengWu optimizes its predictors in a region-adaptive manner. This adaptive approach not only improves the accuracy of individual forecasts but also contributes to the model's overall robustness and reliability. Furthermore, FengWu implements a replay buffer mechanism, which serves to augment its medium-range forecast performance. FengWu refines its predictions over time using historical data and past experiences. In other words, it is continuously learning and adapting to evolving environmental conditions. In terms of performance, FengWu outperforms GraphCast in predicting 80 % of the reported predictands based on the 6-hourly weather forecasting in 2018. It reduces the RMSE of a 10-day lead global geopotential at 500 hPa prediction from 733 to 651 m^2s^{-2} . A sample of forecasted results from FengWu is shown in Fig. 5.

FengWu achieves a skillful forecast lead time of 10.75 days for geopotential at 500 hPa, which is better than GraphCast. The development of FengWu was carried performed on a cluster of 32 NVIDIA A100 GPUs over a period of 17 days. In contrast, GraphCast was trained using 32 Cloud Tensor Processing Unit (TPU) v4 units for 21 days. It has been noted by Jouppi et al. (2023) that TPU v4 chips operate approximately 1.2–1.7 times faster than NVIDIA A100 GPUs. Therefore, the training time for GraphCast, when converted to the equivalent time on NVIDIA A100 GPUs, is estimated to be between 25 and 36 days. Since the training of FengWu was completed in 17 days on GPUs, it represents approximately 47 %–67 % of the equivalent GPU training time for GraphCast, indicating that FengWu was trained faster relative to GraphCast after accounting for hardware differences.

6.2. ClimaX

The ClimaX model uses novel architecture based on the ViT architecture, token embeddings, and large-scale meteorological datasets to capture intricate spatiotemporal relationships within atmospheric data (Nguyen et al., 2023b). It demonstrates remarkable accuracy and generalizability across diverse meteorological parameters and lead times. The ViT architecture incorporates attention mechanisms crucial for capturing long-range dependencies and spatial relationships within meteorological data. ClimaX follows a two-phase training approach, achieving superior performance in long-term and lead-time-conditioned forecasting. An example of the forecast obtained from ClimaX is shown in Fig. 6.

Despite lower spatial resolutions, ClimaX outperforms other models in forecasting the geopotential at 500 hPa, temperature at 850 hPa, and the U component of wind at 10 m above the surface beyond 3 days, in terms of both RMSE and ACC. Even in predicting the U component of wind at 10 m at 7-day lead times, ClimaX surpasses other methods due to its fine-tuned direct prediction based on the mitigating error accumulation for long horizons. While Pangu-Weather performs better on most tasks initially, the gap diminishes with increased lead time, and ClimaX even outperforms Pangu-Weather in predicting the U component of wind at 10 m in 7 days.

6.3. Pangu-Weather

Pangu-Weather is an innovative model based on the relationship between meteorological data and 3D NN (Bi et al., 2023). The Pangu-Weather model first organizes data into smaller blocks and combines them into a 3D cube. It uses Swin encoding, which utilizes a

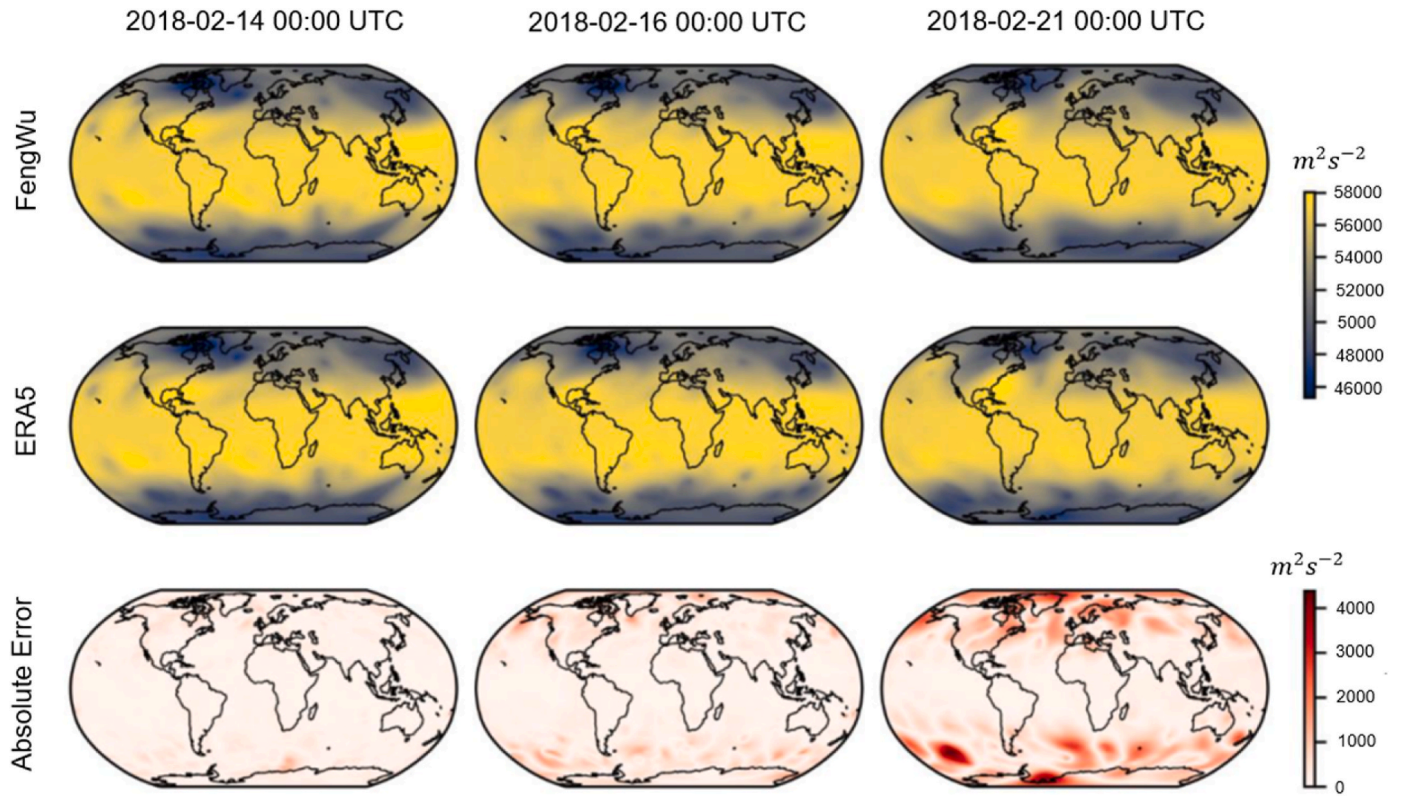


Fig. 5. Example of global map for geopotential at 500 hPa from FengWu and ERA5. Initialization time is 2018-02-11 00:00 UTC (Chen et al., 2023b).

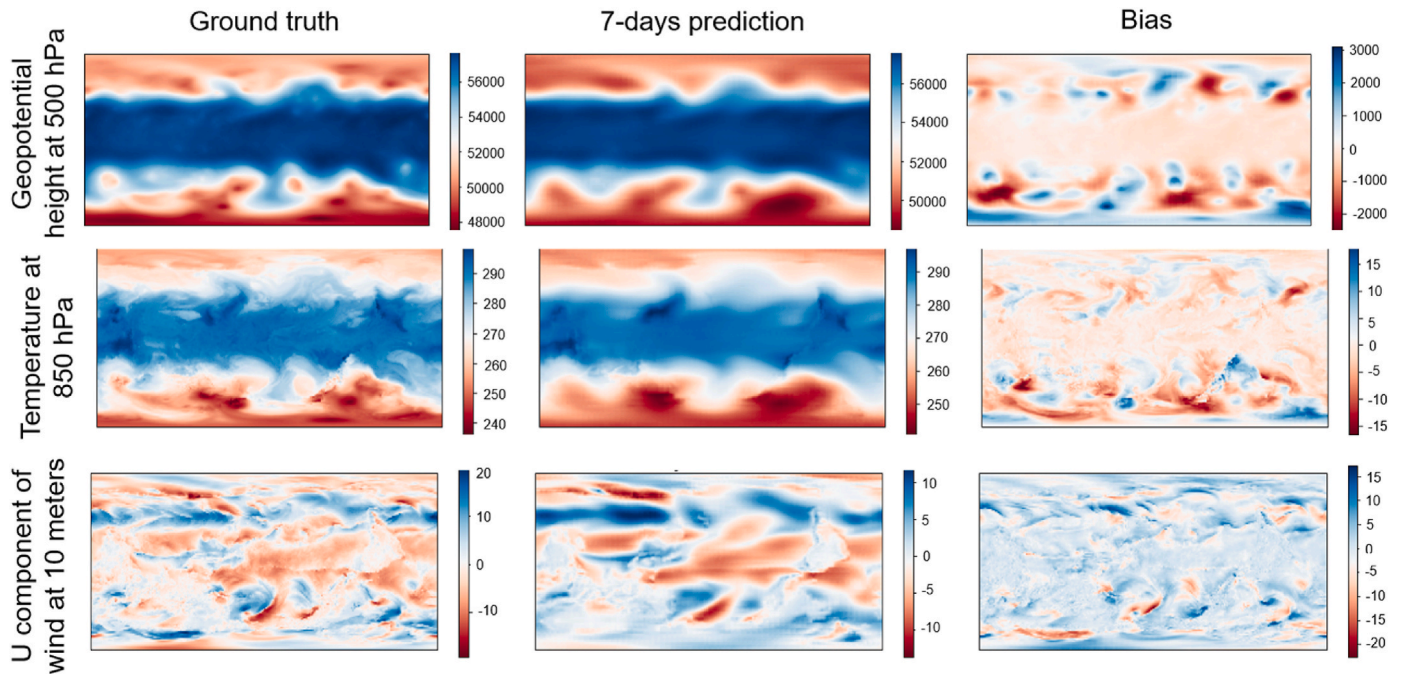


Fig. 6. Sample of the forecasts compared to ground truth ERA5 data (Nguyen et al., 2023b).

self-attention mechanism for comprehensive data processing. After encoding, the model decodes the data, splits it into atmospheric and surface variables, and restores its original resolution for analysis. Additionally, Pangu-Weather introduces a hierarchical time-aggregation strategy, consistently invoking the network with the maximum lead time, thus reducing errors compared to multiple runs of models like FourCastNet. A sample of obtained forecast from this model

is shown in Fig. 7. Moreover, Pangu-Weather shows an ability to accurately predict typhoon trajectories 48 h earlier than NWP. This model is significantly faster than the operational IFS, enabling large-member ensemble forecasts with minimal computational costs. Pangu-Weather is almost 10,000 times faster than the IFS and comparable to the speed of the FourCastNet.

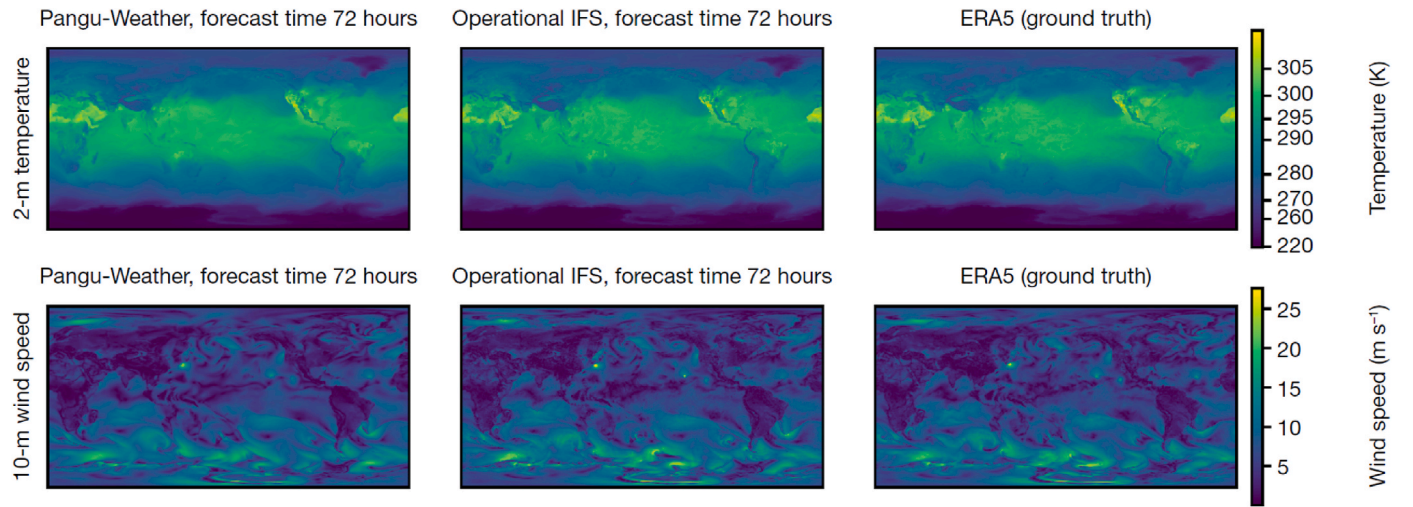


Fig. 7. Visualization of forecast results. The 3-day forecast variables obtained (Bi et al., 2023).

6.4. FourCastNet

FourCastNet uses AFNO for high-resolution inputs, showcasing notable strides in ML (Pathak et al., 2022). AFNO merges the FNO learning method with ViTs' self-attention mechanism. While FNO effectively handles periodic data and complex partial differential equation systems, its computational demands for high-resolution inputs present challenges. Hence, AFNO employs the Fast Fourier Transform (FFT) in the Fourier domain for continuous global convolution. The developers of FourCastNet have mentioned that it is 45,000 times faster than NWP models on a node-hour basis (Pathak et al., 2022). This substantial acceleration, along with its unparalleled accuracy, enables the cost-effective generation of extremely large ensemble forecasts, thereby revolutionizing probabilistic weather forecasting. Fig. 5 shows the example of a prediction obtained from FourCastNet. This model has

the ability to produce massive ensemble forecasts for events such as hurricanes and atmospheric rivers in the order of seconds. Notably, the energy required to train FourCastNet is equivalent to that needed to generate a 10-day forecast with 50 ensemble members using the IFS model. However, once trained, FourCastNet consumes approximately 12,000 times less energy to generate forecasts than the IFS model. Comparatively, FourCastNet matches the accuracy of the IFS model up to three days in advance and shows potential for further improvement. Fig. 8 indicates that the model accurately tracks Super Typhoon Mangkhut's formation and progression over four days (Inset 1). Additionally, the forecast notes three hurricanes forming in the Atlantic Ocean, heading towards North America's eastern coast (Inset 2).

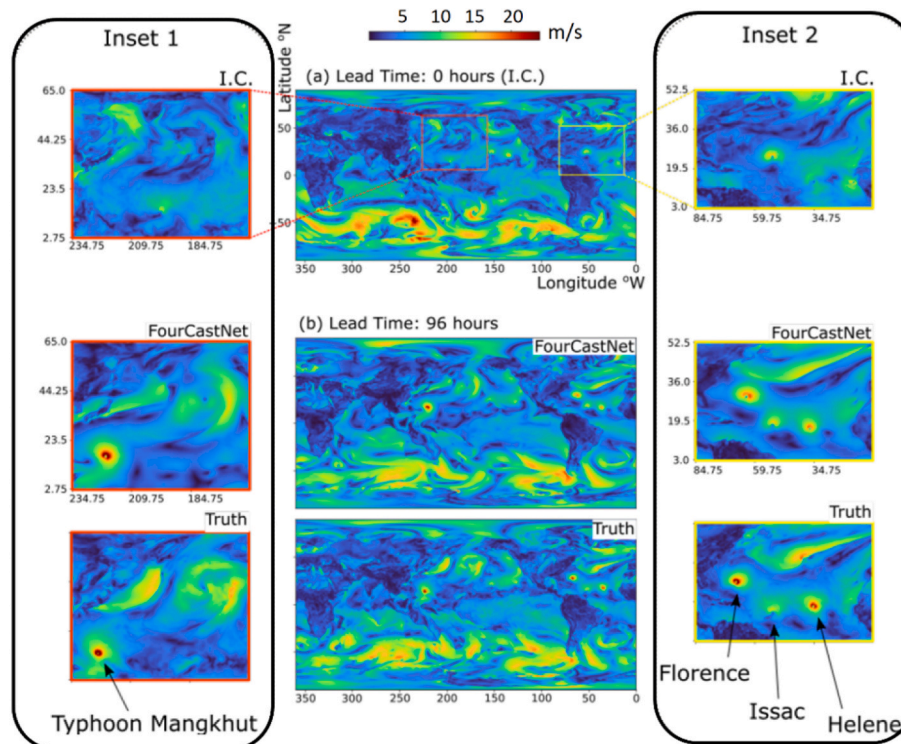


Fig. 8. Example of a global near-surface wind prediction by FourCastNet (Pathak et al., 2022).

6.5. GraphCast

GraphCast is a groundbreaking advancement in weather forecasting, integrating DL with intricate dynamical system modeling based on an autoregressive model structure and a multi-scale mesh representation (Lam et al., 2023). The architecture of GraphCast is shown in Fig. 9. It uses an encode-process-decode framework based on the GNNs to produce forecast trajectories independently. Its encoder component maps the input data onto a multi-grid graphical representation. Subsequently, the processor conducts multiple rounds of message-passing on the multi-mesh. The decoder then translates the multi-mesh representation back to the latitude-longitude grid, generating predictions for subsequent time steps. Experimental results demonstrate GraphCast's superior performance in tracking weather patterns, notably surpassing NWP across various forecasting horizons. However, GraphCast faces challenges in handling uncertainty, primarily focusing on deterministic forecasts. GraphCast's deterministic approach may limit its applicability for longer-term forecasts beyond 10 days. Fig. 10 illustrates examples of statistical parameters derived from GraphCast's evaluation.

6.6. GenCast

GenCast is a powerful probabilistic weather forecasting model that produces ensemble forecasts up to 15 days ahead at a global resolution of 0.25° (Price et al., 2025). At its core is a conditional diffusion model, where the denoising network is divided into three main components: an encoder, a processor, and a decoder. The encoder maps the input data, which includes the noisy atmospheric target and its conditioning history, from a standard latitude-longitude grid to a more efficient icosahedral mesh. This transformation is crucial for preserving spatial coherence and enabling more effective computation. The processor is a graph-transformer that operates on this mesh. It allows each node to attend to its local neighborhood and capture complex spatial relationships through attention mechanisms. Finally, the decoder projects the processed internal representation back onto the original grid, producing a denoised and physically consistent forecast state. This architecture not only allows GenCast to model high-dimensional spatiotemporal dependencies, but also helps maintain the realism and sharpness of its output. In terms of accuracy, GenCast surpasses the leading the

ensemble forecast of the ECMWF in nearly all standard metrics for ensemble forecasting. Its ensemble forecasts also outperform others in applications such as tropical cyclone tracking, providing more accurate mean paths and more informative probability fields. Fig. 11 shows a sample of the outputs generated by GenCast.

6.7. AIFS

The AIFS utilizes an architecture that combines GNN encoders and decoders with a sliding window transformer processor. The system's modular and flexible design supports various levels of parallelism, allowing it to handle high-resolution input data efficiently. Unlike many other models, AIFS operates on a reduced Gaussian grid that maintains uniform spatial resolution worldwide, reducing the number of grid points and edges required and improving computational efficiency. Early versions used an icosphere-based processor grid, but recent iterations have shifted to an octahedral reduced Gaussian grid and adopted transformer-based attention mechanisms for improved performance. The attention mechanism is carefully designed to consider neighborhood relationships along latitude bands, enabling the model to capture spatial dependencies effectively without explicitly using edge features. Regarding accuracy, AIFS often outperforms traditional physics-based models such as ECMWF's IFS, particularly after the first day of forecasts, showing significant improvements in forecast errors throughout the troposphere and at the surface. While some limitations remain, such as reduced skill in the stratosphere and a tendency for forecast smoothing or blurring over longer lead times, the model's performance in capturing tropical cyclone tracks is notably superior to conventional models. Fig. 12 shows a sample comparison between AIFS, GraphCast, and IFS.

Recently, through partnerships with national meteorological services across Europe, ECMWF has developed Anemoi, an open-source Python framework focused on AI-based weather forecasting. Anemoi's goal is to make weather prediction technologies more accessible. It offers key functionalities for training SOTA AI models like AIFS, supports their operational deployment, and helps integrate them into forecasting systems. The framework is made up of several modules, including dataset management, flexible training tools, model deployment utilities, and efficient inference capabilities for real-time use. Moreover, Anemoi

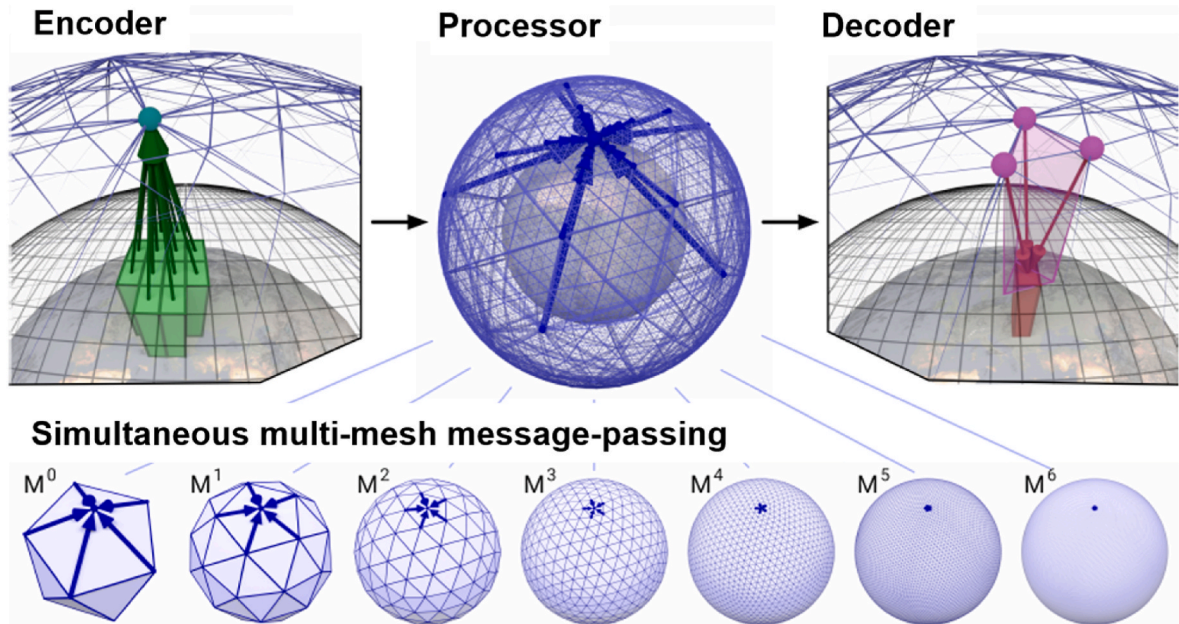


Fig. 9. GraphCast model process involves encoding input regions, updating nodes through message-passing, and decoding processed features back to the grid. The multi-mesh enables uniform global coverage with simultaneous message-passing across edges for comprehensive node updates (Lam et al., 2023).

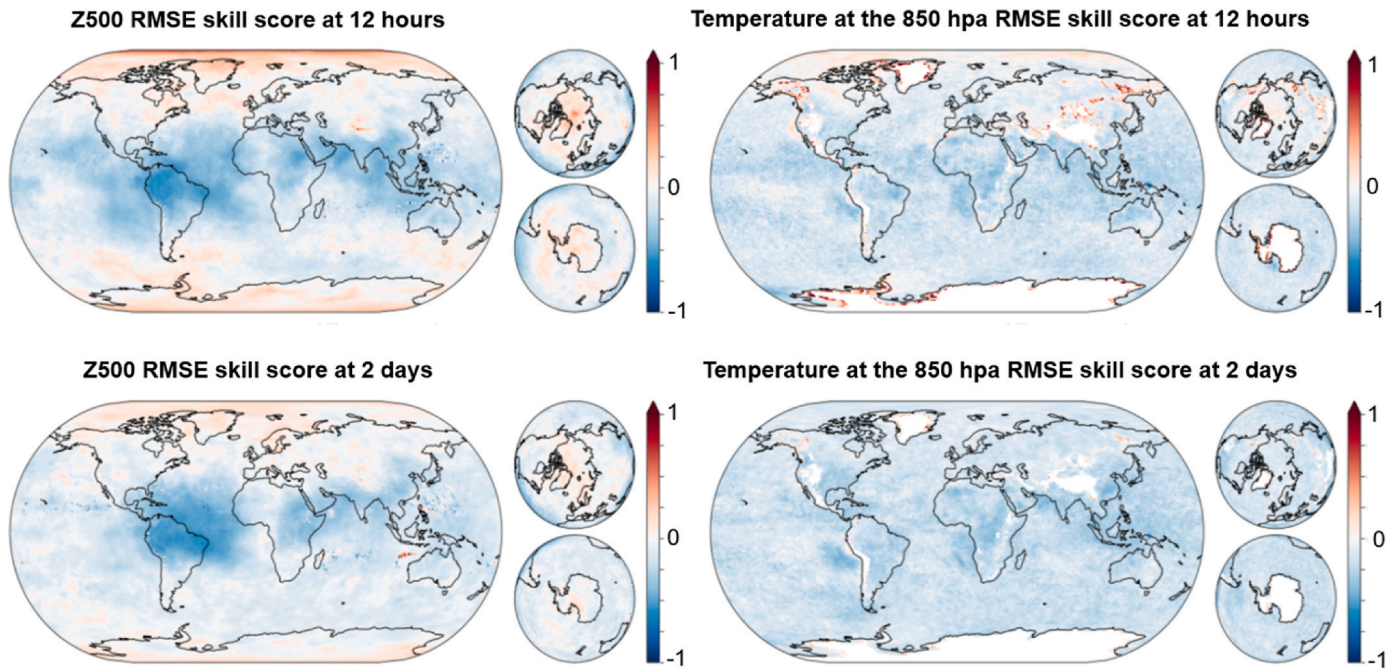


Fig. 10. RMSE difference between GraphCast and HRES, indicating regions where GraphCast outperforms HRES (blue) or vice versa (red) (Lam et al., 2023).

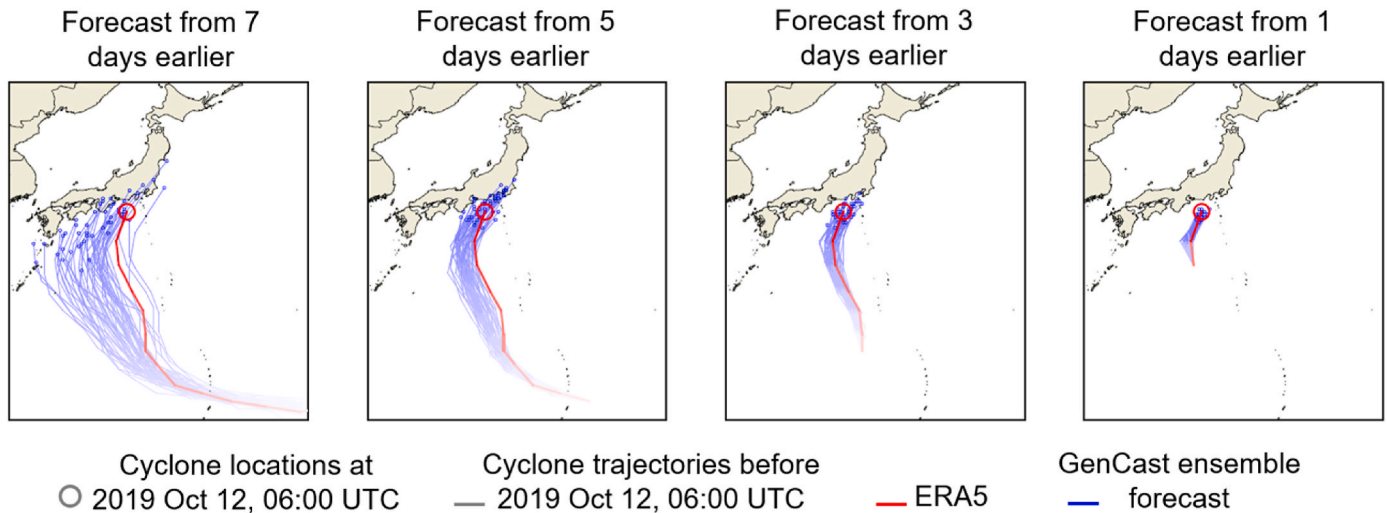


Fig. 11. The path of a typhoon depicted using ERA5 data alongside the ensemble of tropical cyclone tracks from GenCast, shown up to 4 h before the cyclone's landfall in Japan (Price et al., 2025).

Graphs enables experimentation with novel graph architectures, encouraging research and innovation. Designed to promote collaboration, this framework invites contributions from meteorological institutions worldwide, fostering a global community effort.

6.8. Operational comparison of the models

Advanced models like FourCastNet are extremely fast, which reduces the time and cost of large-scale weather forecasting. This speed is crucial because it allows scientists to run many forecast simulations quickly, helping them better understand and evaluate forecast uncertainty. FengWu is one of the models that actively addresses these uncertainties. Its mechanism for mitigating uncertainty loss is a well-designed approach, especially for models sensitive to inherent forecast variability, particularly in long-term predictions. Additionally, GenCast has made significant strides in tackling uncertainties and extreme weather

prediction by providing ensemble-based weather forecasting. While several studies report performance comparisons among SOTA models, there is no comprehensive evaluation covering all models under identical conditions. Instead, existing comparisons often focus on limited subsets of models. To provide insight, this review compiles performance results reported across different studies.

A comparison between FengWu and GraphCast conducted by Chen et al. (2023b) indicates that FengWu generally offers improved long-term performance, as reflected in lower RMSE scores. This enhancement typically does not come at the expense of short-term accuracy, with the exception of the 2-m temperature variable.

According to evaluations using the WB dataset, FuXi demonstrates lower RMSE values than GraphCast at extended lead times for several variables, including sea-level pressure and 2-m temperature (Chen et al., 2023a). However, these gains are not consistent across all scenarios. For instance, short-term FuXi exhibits reduced accuracy in forecasting 2-m

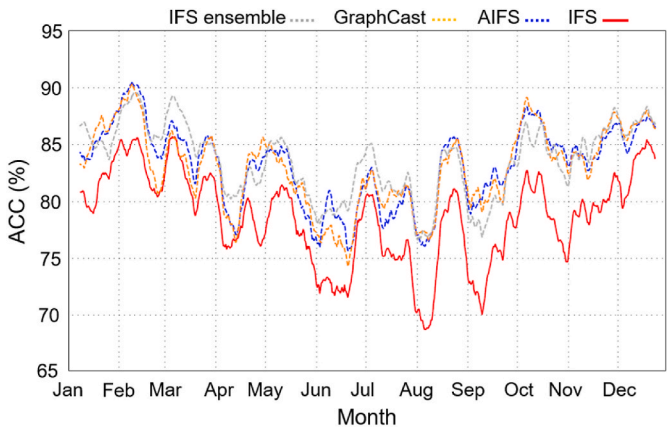


Fig. 12. ACC for AIFS, IFS, GraphCast, and the IFS ensemble forecast mean of 30-day running means of 7-day forecasts at 500 hPa in the Northern extra-tropics during 2022 (Lang et al., 2024).

temperature compared to GraphCast. Interestingly, spectral analysis reveals that FuXi generates forecasts with higher power in this variable and time range, which suggests more detailed but less accurate predictions. A similar trend is observed for 10-m wind speed, where long-term FuXi Long underperforms relative to GraphCast.

Nguyen et al. (2023b) found that ClimaX, despite being trained on significantly lower-resolution data, surpasses FourCastNet in forecasting variables such as 500 hPa geopotential height, temperature at 850 hPa, and wind speed at 10 m at lead times of three days or more, based on both RMSE and ACC metrics. For 2-m temperature, ClimaX delivers improved performance at forecast horizons exceeding three days. While PanguWeather generally achieves stronger results than ClimaX across most variables, the performance gap between the two narrows as the lead time increases. Notably, ClimaX even outperforms PanguWeather in predicting wind speed at 10 m at a 7-day lead time. This advantage is attributed to ClimaX’s fine-tuning for direct prediction, which helps mitigate error accumulation over longer forecast horizons. Although GraphCast records the lowest RMSE overall, its ACC scores fall short compared to those of ClimaX and PanguWeather.

Lang et al. (2024) convincingly demonstrated that AIFS performs far more similarly to GraphCast than to the ECMWF IFS, highlighting the effectiveness of their shared data sources and comparable training strategies. Both AIFS and GraphCast substantially outperform IFS in predicting tropical cyclone positions, achieving markedly lower forecast errors. This enhanced accuracy is primarily due to a significant reduction in the slow bias of tropical cyclone propagation speed, which directly contributes to improve along-track forecast performance.

Several studies have evaluated the performance of various models across specific regions and events (Liu et al., 2024; Pasche et al., 2025). Liu et al. (2024) examined model accuracy in forecasting several typhoons that impacted Eastern Asia and the Western Pacific between June and November 2023. In this context, Pangu-Weather demonstrated the largest track errors, with errors decreasing progressively for FourCastNet, GraphCast, FuXi, and FengWu up to 144 h. However, at the 168-h mark, FengWu’s track error surpasses FuXi’s and is higher than GraphCast’s. In addition they analyzed RMSE and ACC for multiple variables such as 500 hPa geopotential height, temperature at 850 hPa and 2 m, and the zonal wind component at 10 m. Their findings indicate that FengWu consistently achieves the lowest RMSE, followed by FuXi and GraphCast, while FourCastNet and Pangu-Weather rank lower in performance. Furthermore, Pasche et al. (2025) conducted a detailed comparison of GraphCast, Pangu-Weather, and FourCastNet across three significant weather events: the 2021 Pacific Northwest heatwave, the 2023 South Asian humid heatwave, and the 2021 North American winter storm. During the 2021 Pacific Northwest heatwave, forecasts from Pangu-Weather and GraphCast were on par with HRES based on

the evaluated metrics. However, for shorter lead times, HRES produced lower forecast errors than the AI-based models, a reversal of trends seen during the baseline summers of 2020 and 2022. This suggests that AI-based models may struggle more when extrapolating to extreme weather conditions. Interestingly, HRES faced greater challenges than the AI-based models in capturing the rapid temperature decline following the heatwave’s peak. In the case of the South Asian humid heatwave, the AI-based models tended to underestimate the highest risk zones over Bangladesh, reflecting spatially distinct error patterns. For the North American winter storm, Pangu-Weather and GraphCast frequently delivered more precise forecasts than HRES across many lead times and initial conditions. The study also identified fundamentally different error dynamics: HRES and FourCastNet appeared more sensitive to subtle variations in initial conditions, resulting in error accumulation before the events, whereas GraphCast and Pangu-Weather were less affected.

Regarding the hardware used to run the models, Table 10 summarizes the computational resources for each. Pangu-Weather relies on large GPU/TPU clusters over extended training periods, emphasizing model complexity and scale. In contrast, FourCastNet and FengWu focus on fast inference with fewer resources. GenCast and GraphCast utilize TPU cloud infrastructure for scalable parallelism, while AIFS balances training and inference efficiency with optimized GPU usage. The models vary widely in hardware requirements, training durations, and inference speeds.

7. Discussion

One of the advantages of AI-based weather forecasting models is their computational efficiency. They could be trained using accessible hardware like GPUs, TPUs, or specialized AI hardware, while NWP models need HPC clusters. Speed is another notable advantage of these models. They can forecast weather phenomena significantly faster than NWP models, with latencies typically in the order of seconds. However, it’s important to note that while AI-based models may offer faster processing, they might not resolve as many physical processes as higher-resolution NWP models. Despite this, AI-based models remain independent of certain resolution constraints and offer advantages in terms of computational efficiency and speed.

Table 10
Computational resources and performance characteristics.

Model	Computational resources
FengWu	FengWu was developed using the PyTorch framework and trained on a cluster of 32 NVIDIA A100 GPUs over 17 days. Its inference speed on an NVIDIA Tesla A100 GPU is under 30 s to generate forecasts for the next 10 days at 6-h intervals.
ClimaX	ClimaX employed up to 80 NVIDIA V100 GPUs (32 GB each) for training
Pangu-Weather	Each network in Pangu-Weather was trained for 100 epochs, requiring about 16 days on a cluster of 192 NVIDIA Tesla V100 GPUs.
FourCastNet	FourCastNet runs on a single node of the Perl mutter High-Performance Computing (HPC) cluster, which contains 4 A100 GPUs per node, and can produce a 100-member 24-h forecast in 7 s.
GraphCast	Training GraphCast took roughly 21 days on 32 cloud TPU v4 devices using batch parallelism.
GenCast	GenCast operates on Cloud TPU v5 devices, generating a single 15-day forecast in about 8 min. It can also produce ensembles of forecasts in parallel by utilizing cloud scalability and parallel computation.
AIFS	AIFS trains on 64 NVIDIA A100 GPUs (40 GB each) with data parallelism and mixed precision, using four GPUs per node. Training lasts about a week, and inference for a 10-day forecast takes 2.5 min on a single A100 GPU.

7.1. Challenges

Several challenges persist in AI-based weather forecasting models in achieving optimal performance and accuracy.

7.1.1. Limited, low-quality, low resolution data and small-scale forecasting

The scarcity of high-quality, high-resolution historical meteorological data poses a significant obstacle to the effective application of AI in weather forecasting. Limited and low-quality datasets reduce the reliability of AI models by undermining both the training process and the validation of forecasting algorithms. The success of AI-based models fundamentally depends on training data and initial conditions that are diverse and accurate. Biases and inaccuracies often arise from uneven data coverage or changes in measurement techniques over time and space. For example, historical meteorological records may contain inconsistencies due to evolving sensor technologies or shifting environmental conditions. These factors diminish the representativeness of training data. Therefore, ensuring that AI-based models are trained on data that are high-quality, diverse, and representative is essential to reducing errors and improving forecast reliability.

Current SOTA AI-based weather models primarily rely on nearly four decades of ECMWF ERA5 data for both training and validation. For single-step forecasting, these models use ERA5 data as the initial condition. In multi-step forecasting, the process begins with ERA5 data at the initial time. The model then recursively uses its own previous predictions, which are generated at the same 0.25° (~ 30 km) resolution, as input for subsequent time steps. This autoregressive approach represents a significant advancement. Nevertheless, the relatively coarse resolution of ERA5 limits the models' ability to resolve small-scale atmospheric phenomena, particularly convection. Accurately predicting convection remains a major challenge because the ~ 30 km resolution is insufficient to capture convective processes, which are key drivers of severe weather events such as thunderstorms (Gustafsson et al., 2018; Buschow et al., 2024). Without adequate resolution, forecasts struggle to represent small-scale dynamics that influence localized weather outcomes. This limitation highlights the urgent need to incorporate high-resolution, high-quality humidity data, at least in initial conditions, and ideally throughout all training and validation stages. Such data are critical for improving small-scale weather forecasts.

Some studies have applied downscaling techniques to enhance the spatial resolution of forecast results (Nguyen et al., 2023b; Liu et al., 2024). However, research indicates that simply increasing spatial resolution does not necessarily improve model performance when critical parameters like humidity remain inaccurate (Ducrocq et al., 2002). Therefore, improving humidity data accuracy in ERA5 through assimilation of high-quality, high-resolution tropospheric observations from advanced technologies such as remote sensing techniques is essential. Integrating these data will enhance key parameters needed for convective-scale modeling and ultimately lead to better AI-based weather forecasts.

Current AI models typically rely on NWP outputs and their data assimilation pipelines, limiting their ability to fully replace conventional forecasting. Emerging data-driven models, such as Aardvark system, demonstrate promising results by learning direct mappings from raw observational data to forecasts, thereby reducing computational costs and complexity. However, these models face challenges including underperformance in predicting extreme weather events, reliance on ERA5 data for training, and the lack of standardized evaluation benchmarks.

7.1.2. Explainability

AI-based models, particularly those used in weather forecasting, often face significant challenges related to explainability. They frequently function as "black boxes" with limited transparency. Understanding complex weather dynamics is essential for accurate forecasting. eXplainable AI (XAI) methods such as SHapley Additive exPlanations

(SHAP), Local Interpretable Model-agnostic Explanations (LIME), and Gradient-weighted Class Activation Mapping (Grad-CAM) help improve transparency by clarifying how AI models generate their predictions (Bhattacharya, 2022; Yang et al., 2023; Van Zyl et al., 2024). These techniques enable meteorologists to identify patterns, evaluate correlations, and assess the relative influence of input variables. The opaque nature of many AI-based forecasting models raises concerns among meteorologists about their reliability and trustworthiness. When it is unclear how meteorological factors contribute to predictions or why certain results are produced, it becomes difficult to evaluate outputs critically or detect faulty correlations. XAI techniques, when used alongside standard statistical evaluation methods, provide an opportunity to demonstrate the reliability of AI-based weather models. By offering logical and verifiable explanations, XAI allows experts to validate model predictions against established meteorological knowledge, which fosters greater confidence in AI-based forecasts.

7.1.3. Physical constraints

While many studies have demonstrated the advantages of AI techniques over traditional modeling and statistical techniques in scientific applications, concerns persist regarding the reliability and robustness of the model, particularly in preserving fundamental physical principles. In unfamiliar scenarios or outside the training domain, the model models may deviate from established physical laws, which limits their generalization capabilities. To address these challenges, Knowledge-Guided Machine Learning (KGML), also known as Physics-Informed Machine Learning (PIML), has emerged as a promising approach. This method integrates physical laws and domain knowledge into ML or DL models to enhance both accuracy and interpretability. For instance, Beucler et al. (2019) showed that applying physical constraints improves the performance and relevance of the model in climate modeling. Similarly, studies by Wu et al. (2020) and Daw et al. (2020) demonstrated that incorporating physics-based constraints allows more effective simulation of complex systems such as turbulence and Earth's climate. This results in models that exhibit greater consistency, improved generalizability, and better uncertainty estimation. One specific and increasingly popular approach within PIML is the use of Physics-Informed NNs (PINNs). These models embed physical laws, often in the form of Partial Differential Equations (PDEs), directly into the loss functions of NNs. By penalizing deviations from known physical behaviors during training, PINNs ensure that the resulting models not only fit data but also respect underlying scientific principles. This is particularly advantageous in domains with limited or noisy data, where physical laws can serve as a form of regularization and guidance. Integrating physical information into AI-based forecasting models improves both prediction accuracy and consistency with established physical relationships. This is especially critical in meteorology, where variables such as temperature, pressure, and humidity are closely interdependent. PIML methods, including PINNs, enable the incorporation of these interactions into the modeling process, which is essential for high-stakes applications like extreme weather forecasting. Despite progress in AI-based weather forecasting, challenges remain in modeling multiple meteorological parameters simultaneously while capturing their nonlinear interactions. The highly dynamic nature of the atmosphere means that variations in one parameter can substantially influence others. Therefore, a more comprehensive modeling strategy is necessary. The integration of PIML, particularly PINNs, with conventional techniques offers a promising pathway toward achieving physically consistent and scientifically credible predictions.

7.1.4. Uncertainties and extreme weather prediction

Weather forecasts are inherently uncertain due to the complex and chaotic nature of the atmosphere. This uncertainty requires predicting a range of possible weather scenarios instead of relying on a single deterministic forecast. Recent AI-based weather models have improved forecast accuracy but mainly focus on deterministic predictions and

often lack reliable uncertainty quantification. The GenCast model addresses this limitation by providing probabilistic forecasts that are both faster and more skillful than leading operational ensemble systems. This method captures forecast uncertainty effectively and improves decision-making in weather-sensitive fields. However, probabilistic approaches like this are not yet widely adopted in other AI-based models. One important benefit of ensemble-based weather forecasting is the opportunity to increase the accuracy of extreme weather predictions. Evaluation of AI-based models in extreme weather prediction remains incomplete. Despite progress in forecasting specific extremes such as hail size (Billet et al., 1997), tornadoes (Adrianto et al., 2009), and extreme precipitation (Radhika and Shashi, 2009; Herrmann and Schumacher, 2018), and improvements in forecasting tropical cyclone tracks and wind power production with GenCast, significant challenges still persist. Meteorologists also identify prediction of out-of-distribution severe weather events as a major problem for AI-based forecasting. In addition, as mentioned earlier, the quality and resolution of input data, especially initial conditions, are often insufficient for predicting convective-scale processes, which are key to many extreme weather events. Therefore, increasing the reliability and resolution of critical parameters such as humidity in the initial conditions could significantly enhance forecast skill.

7.1.5. Temporal adaptation and generalization

Given the dynamic nature of atmospheric systems, forecasting models must continuously adapt to accurately capture changing weather patterns. Without mechanisms to address temporal dynamics, models risk becoming less effective, especially as environmental conditions deviate from historical norms. Ongoing refinement is essential to ensure precise and timely forecasts amid emerging climate trends. Moreover, AI models often face limitations in generalizing to unseen or extreme weather events beyond their original training data. This constraint reduces the robustness and reliability of such models, especially in critical scenarios where accurate prediction is paramount. Addressing these challenges by integrating temporal adaptation and enhancing generalization capabilities is crucial to advancing AI-based weather forecasting and improving its practical utility. Previous research has insufficiently addressed these critical aspects. To advance the field of weather forecasting, it is imperative that these oversights be rectified.

8. Conclusion

As discussed, AI-based weather forecasting has seen significant advancements in recent years, particularly since 2022, marking a milestone in the field due to its remarkable flexibility, speed, and computational efficiency. This paper presents a comprehensive review of AI-based weather forecasting models, highlighting their rapid development and key characteristics. It emphasizes the potential of these models to improve forecasting accuracy. The evolution of AI methods such as RNNs, CNNs, and GNNs, along with the development of models like FourCastNet, GraphCast, GenCast, and more recent data-driven models like Aardvark, demonstrates the growing capability of AI to handle complex forecasting tasks and capture specific weather phenomena. Considering the performance of the models and existing challenges in the field, moving forward, incorporating high-quality, high-resolution data is essential to enhance the accuracy of small-scale weather and extreme event predictions. While downscaling techniques and autoregressive modeling approaches have contributed to progress, further improvements in data, particularly humidity, will be critical for refining forecast precision at convective scales. Additionally, enhancing the explainability of AI models represents a vital direction for future research. By integrating XAI techniques, transparency can be improved, allowing meteorologists and researchers to better understand model outputs, validate predictions, and identify potential sources of error. Another promising avenue lies in embedding physical constraints into AI models through PIML and PINNs. These approaches promote forecasts

that adhere to fundamental meteorological principles, thereby improving both the accuracy and generalizability of models in unfamiliar or extreme scenarios. Further development in uncertainty quantification and probabilistic forecasting is also crucial, as most current AI-based models focus primarily on deterministic predictions. Probabilistic frameworks like GenCast have demonstrated potential for more skillful and reliable extreme weather forecasting, suggesting the need for wider adoption and rigorous evaluation across diverse AI models.

CRedit authorship contribution statement

Saeid Haji-Aghajany: Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Witold Rohm:** Writing – review & editing, Supervision. **Piotr Lipinski:** Writing – review & editing, Supervision. **Maciej Kryza:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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